

Evaluating Housing Accessibility and Spatial Disparities in Shanghai

Submitted by

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Abstract

Housing costs in Shanghai vary substantially across districts, creating uneven access to different parts of the city. While central districts offer greater proximity to employment opportunities, transportation networks, and urban amenities, they also command significantly higher purchase prices and rents. Understanding the spatial distribution of housing costs is therefore important for evaluating housing accessibility and identifying disparities within Shanghai's housing market. This project evaluates housing accessibility and spatial disparities across Shanghai's 16 districts using web-scraped housing listings, historical housing price data, and resident income statistics. Data were collected from *Lianjia* and *Gotohui* using Python-based web scraping techniques and stored in a MySQL relational database for analysis and querying. Through descriptive statistics and visualizations, the study examines differences in housing prices, rental costs, affordability indicators, and long-term price trends across districts. The findings reveal significant spatial inequalities, with central districts exhibiting substantially higher housing prices and rents than peripheral areas, while affordability remains a challenge for average-income residents. To improve access to housing information, a Python-based query system was developed, allowing users to search for properties based on location, budget, size, and metro accessibility.

Keywords: housing affordability, Shanghai, web scraping, spatial disparities, real estate analytics, database systems.

1. INTRODUCTION

1.1. Background

Housing affordability has become a growing concern in major global cities due to rapid urbanization, increasing demand for housing, and rising property prices. Shanghai, as China's largest commercial and financial center, has experienced substantial economic growth and population concentration, resulting in significant pressure on its housing market. While housing serves as a fundamental human need, rising housing costs can reduce accessibility for middle- and lower-income households and contribute to social inequality. Recognizing the importance of adequate and affordable housing, the United Nations included housing accessibility within Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities. Specifically, SDG Target 11.1 seeks to ensure access for all to adequate, safe, and affordable housing and basic services by 2030 (United Nations, 2024).

In Shanghai, housing affordability cannot be understood only through the city's average housing price, because different districts vary greatly in price level, rental cost, transportation access, and living convenience. Therefore, district-level comparison is necessary to better understand the housing market.

This project uses web-scraped housing data to evaluate housing affordability in Shanghai. By collecting current housing listings, rental listings, historical district-level housing prices, and income-related data, we analyze housing price differences, rental patterns, price-to-income ratios, price-to-rent relationships, and long-term trends across Shanghai's 16 districts.

1.2. Objective

This project aims to:

1. Analyze housing price differences across Shanghai districts.
2. Examine rental market patterns across districts.
3. Evaluate housing affordability using price-to-income and price-rent ratios.
4. Identify factors associated with variations in housing prices.

1.3. Project Scope

This project does not include policy analysis, negotiation-level prices, or advanced forecasting. It mainly focuses on describing and comparing the data using visualizations.

2. CONTENTS

2.1. Data Collection

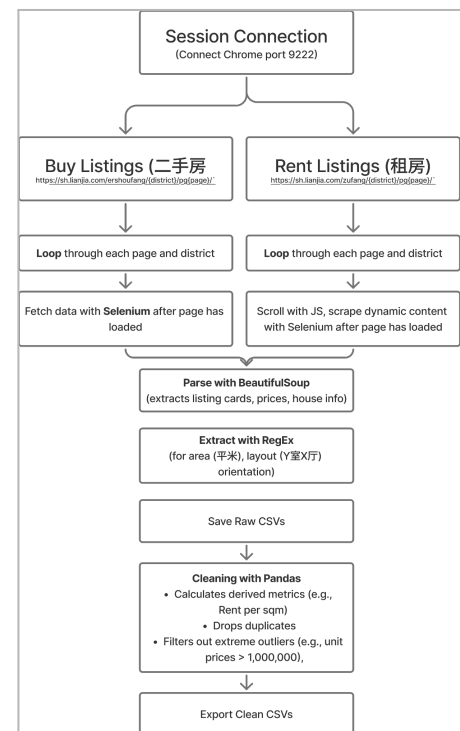
2.1.1. Housing Listings (Lianjia)

Source

Lianjia, one of China's largest online real estate platforms, was used to collect housing listings to obtain current market information on both rental properties and second-hand residential properties across Shanghai districts in June 2026. Each results page on the platform contained approximately 30 housing listings. For data consistency and coverage, we collected data from 5 pages per district. In total, Shanghai's 16 districts were included in the dataset, resulting in approximately 2,000 rental listings and 2,000 second-hand property listings to work with.

Method

The website loads information dynamically and has protections against automated scraping, hence Selenium was used to automate the browsing process and collect listing data. Instead of opening a new browser and logging in each time, the script connected to an existing Chrome browser session through a debugging port (9222). Using the browser's existing login session and cookies makes it easier to access the website without repeatedly signing in. After Selenium loaded the webpage, BeautifulSoup was used to extract the required information from the page's HTML structure. Regular expressions (regex) were used to clean and format text, such as extracting numbers from prices or removing unwanted characters. pandas was used to organize the collected data into tables and prepare it for further analysis.



Challenges

- During data collection, CAPTCHA verification pages appeared when navigating through multiple listing pages.

- Rental listings and second-hand housing listings were organized using different webpage structures.

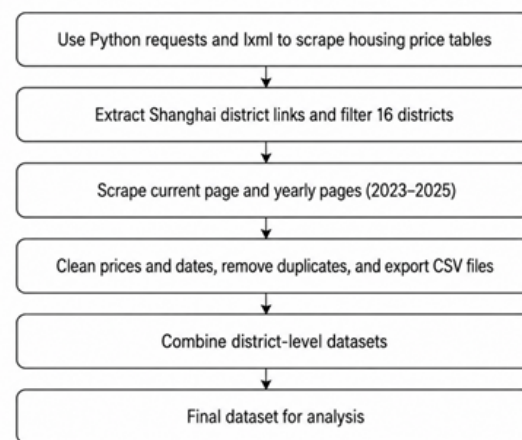
2.1.2. Historical Housing Prices

Source

Historical housing price data were collected from Fangjia Gotohui, a housing price data platform that provides monthly residential price information for different districts in Shanghai. The dataset focuses on Shanghai's 16 administrative districts and covers historical monthly housing prices from 2023 to 2026. The collected variables include the average second-hand housing price and the average new housing price, both measured in yuan per square meter. These historical price records were used to provide a broader market background and support the comparison between current housing listings and long-term district-level price trends.

Method

A Python-based web scraping program was developed to collect the historical housing price data. First, the program automatically accessed the Shanghai housing price data page and extracted links for all district-level pages. Non-specific area links, such as general city-level pages, were excluded to ensure that only valid district-level data were collected.



For each district, the program scraped data from both the current district page and the historical yearly pages for 2023, 2024, and 2025. Since the current page and yearly pages used different table structures, two separate scraping functions were applied. The current page was used to capture recent monthly records, while the yearly pages were used to obtain historical monthly data. After extraction, the data were cleaned by removing extra spaces, Chinese unit symbols, and non-numeric characters from price fields.

To improve data quality, duplicate records were removed based on the date variable. The final dataset only kept records from 2023 to 2026. The cleaned results were then exported into separate CSV files for each district, as well as one combined CSV file containing all Shanghai districts.

Challenges

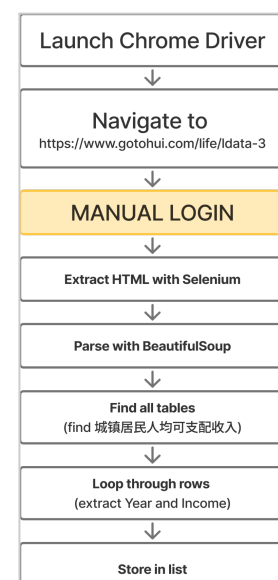
- The website contained both district-level and non-specific area links, so additional filtering was required to keep only valid Shanghai administrative districts.
- The current page and historical yearly pages had different table formats, which required separate scraping logic.
- Housing price values included Chinese text, units, and formatting symbols, so data cleaning was necessary before further analysis.
- Some records appeared repeatedly across current and historical pages, so duplicate observations had to be removed based on the date.
- A short time delay was added between requests to reduce the risk of access failure and avoid sending too many requests to the website in a short period.

2.1.3. Resident Income

Source

Gotohui.com is a Chinese data platform that aggregates various types of economic and social statistics for research, business, and academic use. The data used in this study, specifically the official annual Urban Resident Per-Capita Disposable Income (城镇居民人均可支配收入) for Shanghai, was sourced from this website (www.gotohui.com/life/ldata-3). According to the site, the original data is provided by the Shanghai Statistics Bureau (上海统计局). Urban resident disposable income was selected because it is one of the most commonly used indicators for evaluating housing affordability, as it reflects the amount of income available to households after taxes and mandatory deductions and therefore provides a more realistic measure of residents' capacity to purchase or rent housing.

Method



Since some data is behind a login paywall, it cannot be accessed using standard HTTP requests. For this reason, Selenium was used to control a real Chrome browser, allowing the script to access the website after manual login and retrieve the fully rendered page content for processing. Additional libraries included `webdriver_manager` for managing `ChromeDriver`, `BeautifulSoup` for HTML parsing, the `re` (Regex) library for text extraction, and `csv` for saving the final dataset.

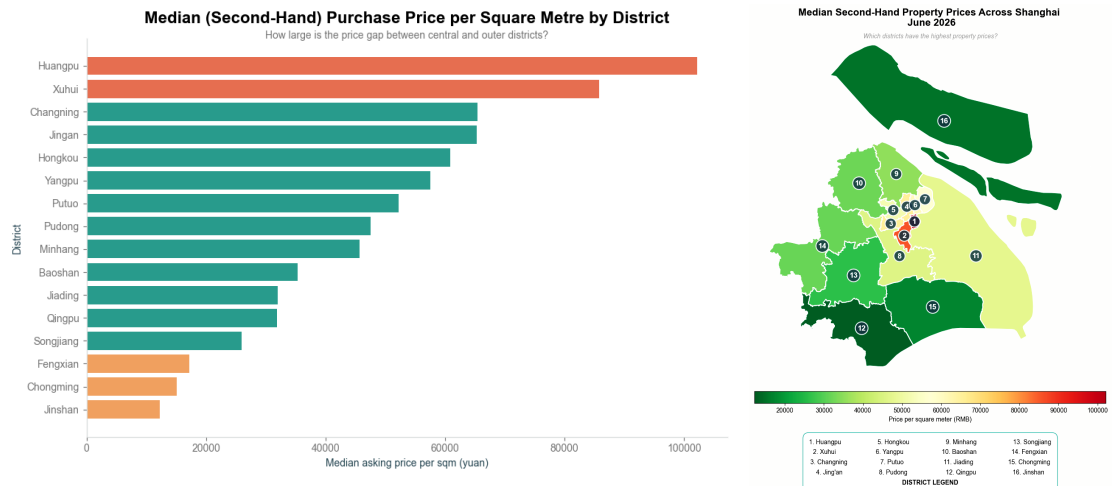
Challenges

- The data is not publicly accessible without an account, so a manual login step was required before scraping could continue. The script was designed to pause during this step, since automating login credentials can trigger security restrictions or block access.

2.2. Data Analysis and Visualization

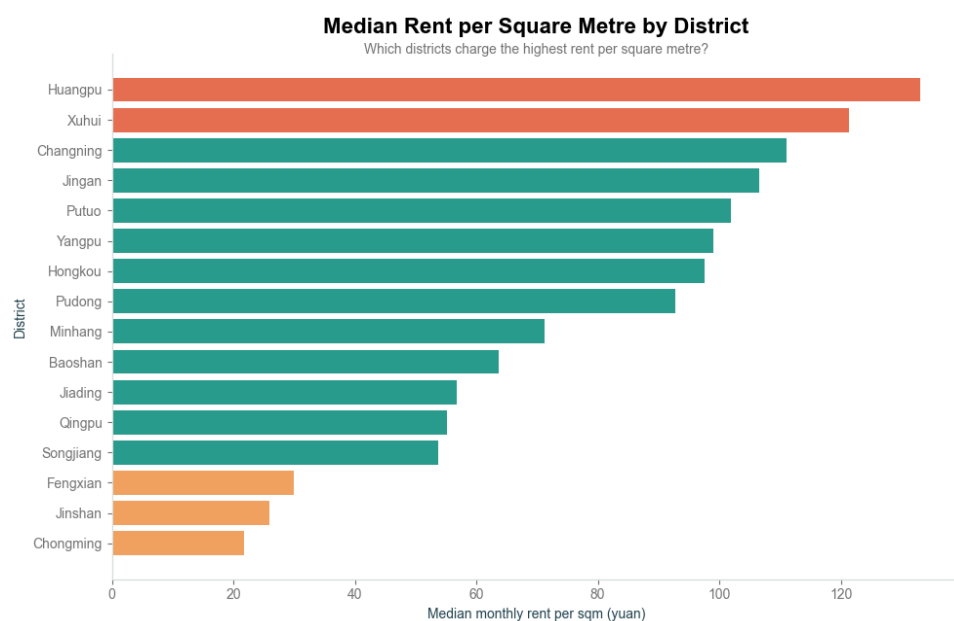
2.2.1. Current Housing Market

- Median Second-Hand Housing Price per Square Metre by District, Shanghai (June 2026)



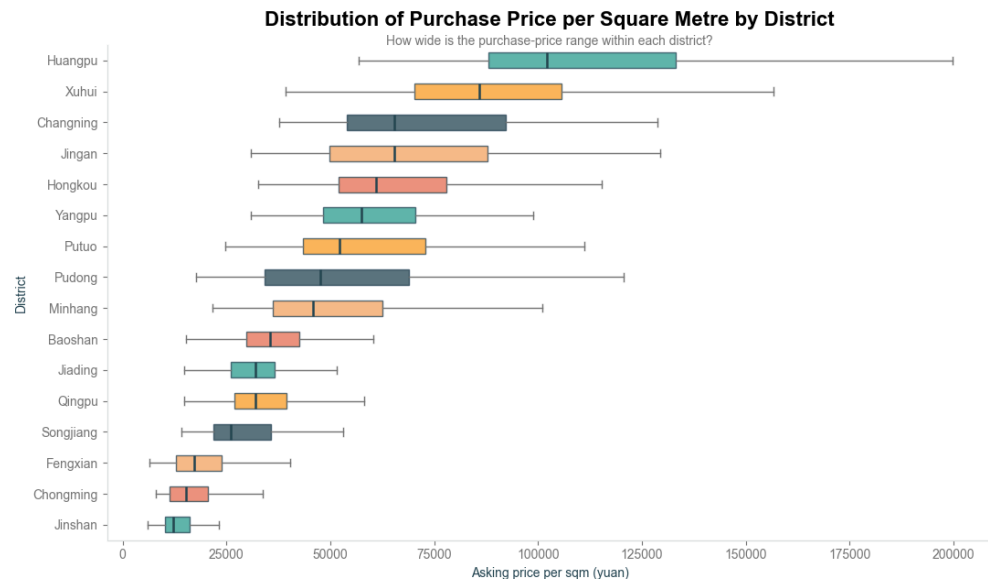
Huangpu had the highest median purchase price at 102,145 yuan per sqm, followed by Xuhui at 85,738 yuan. Jinshan recorded the lowest value at 12,265 yuan, meaning that the Huangpu median was more than eight times higher. This demonstrates a strong price gap between central and outer districts.

- Median Rental Price per Square Metre by District, Shanghai (June 2026)



Rental prices followed a similar geographic pattern. Huangpu recorded the highest median rent at 133.0 yuan per sqm, while Chongming had the lowest at 21.8 yuan. Central-location premiums therefore affect both buyers and renters.

- Distribution of Purchase Price per square metre by District (June 2026)



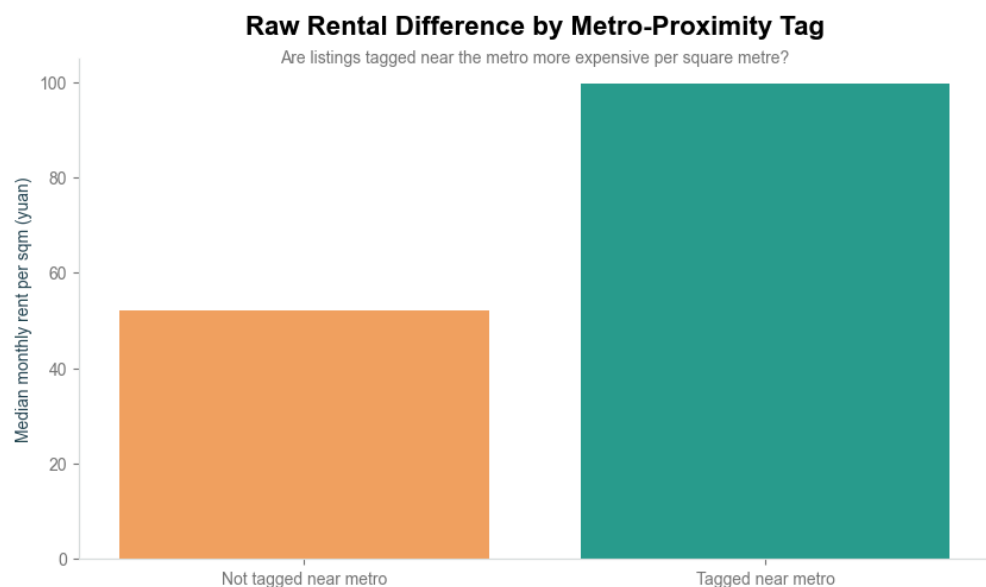
The boxplot shows substantial price variation within several districts, especially Huangpu, Changning, Jingan and Pudong. This means that a district median can hide different market segments, including older housing, standard developments and premium properties.

- Property Area vs (Second-Hand) Housing Price (June 2026)



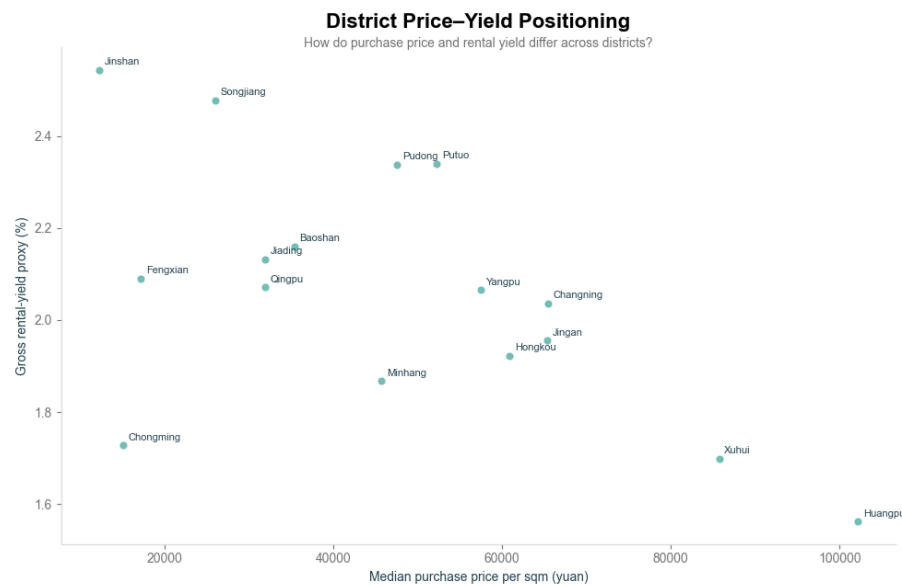
Property area and total asking price had a positive correlation of approximately 0.66. Larger properties generally required higher total budgets, but the wide spread of points shows that location and property characteristics also influence price.

- Rental Price Difference based on Metro-Proximity Tag (June 2026)



Listings marked as near a metro station had a median rent of approximately 99.8 yuan per sqm, compared with 52.0 yuan for other listings. However, the difference may also reflect the concentration of near-metro listings in central and more expensive districts.

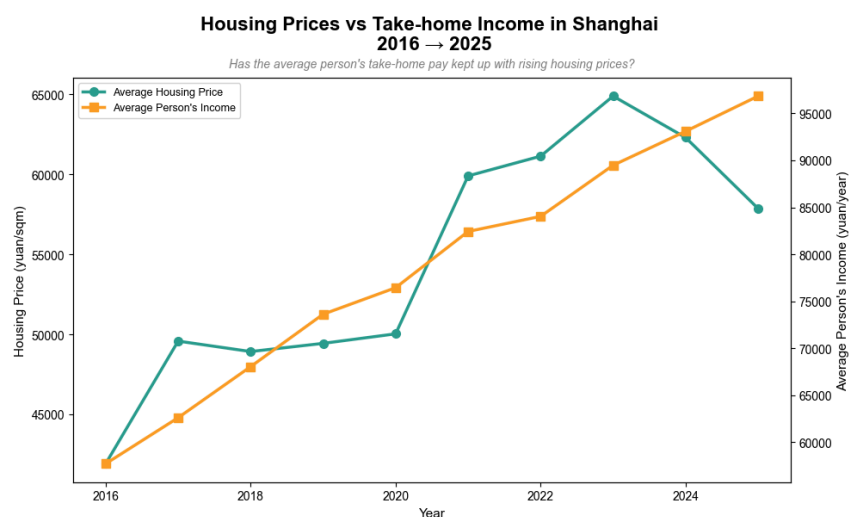
- (Second-hand) Purchase Price vs. Gross Rental Yield by District (June 2026)



Huangpu and Xuhui combined very high purchase prices with relatively low rental-yield proxies of approximately 1.56% and 1.70%. Jinshan and Songjiang produced higher values of approximately 2.54% and 2.48%. Outer districts therefore generated more rent relative to their purchase price, although the calculation excludes expenses and vacancy.

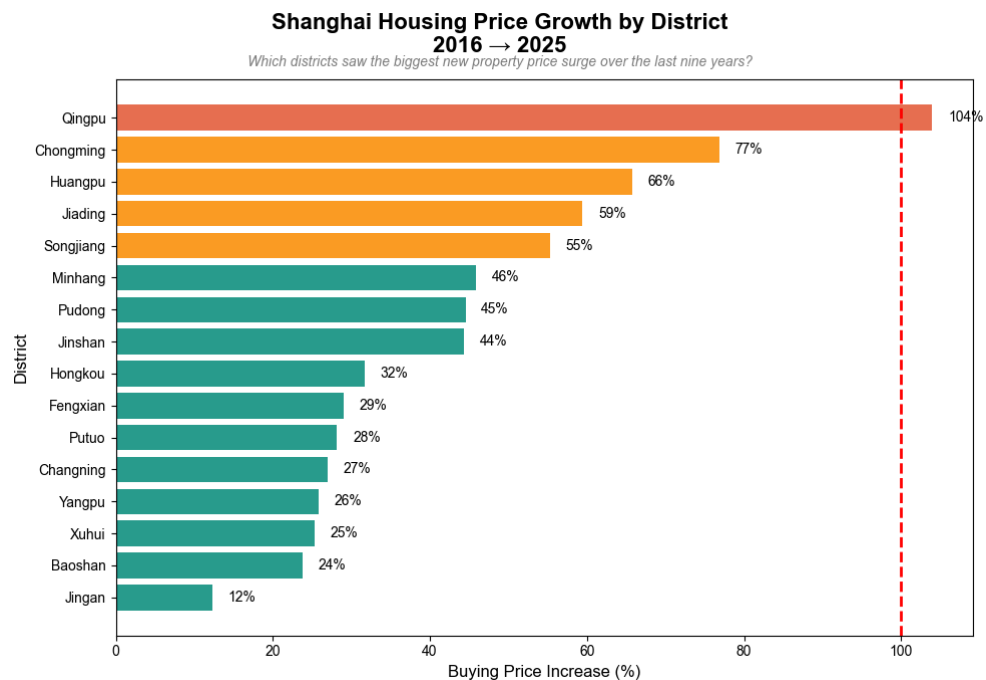
2.2.2. Historical Trend Analysis

- Income Growth vs. Housing Price Growth, Shanghai (2016-2025)



The price spike into 2023 likely reflects post-COVID pent-up demand and speculative buying when restrictions were lifted. Income climbed steadily throughout. By 2025 income has nearly caught up, suggesting affordability is actually improving recently.

- Housing Price Growth Trend by District (2016-2025)



Qingpu is a clear outlier, prices there literally doubled (+104%), likely driven by development spillover from the Yangtze River Delta integration and Hongqiao expansion. Districts within or near the Inner Ring, Jing'an (+12%), Xuhui (+25%), Yangpu (+26%) saw the smallest growth. These areas have been built-out and desirable for decades, hence there isn't much room to expand. Demand is consistently high but supply is extremely constrained.

- Price-to-Income Ratio Trend (2016-2025)



It costs a Shanghai resident roughly 7-9 months of their annual disposable income just to buy 1 sqm of housing. The ratio peaked in 2017 (9.5 months)

and has since trended down to 7.2 months in 2025, which is the most affordable point in a decade.

2.3. Finding Interpretation

2.3.1. Spatial Polarization and Price Tiers

The exploratory data analysis reveals a profound polarization between Shanghai's inner core and its peripheral districts. Based on both purchase and rental distributions, the market can be segmented into three distinct geographic tiers:

- **Inner Ring (Tier 1)**

Districts such as Huangpu and Xuhui exhibit the highest concentration of property values, with median purchase prices exceeding 85,000 Yuan/sqm and rental rates commanding a significant premium. The real estate listings exhibit a slow price growth compared to the other rings.

- **Middle ring (Tier 2)**

Districts like Pudong, Minhang, and Baoshan display moderate pricing but exhibit the widest Interquartile Ranges (IQRs). This wide dispersion indicates highly heterogeneous markets containing both middle-class residential complexes and high-end modern developments. The price growth in these districts can be qualified as moderate to high.

- **Outer ring (Tier 3)**

Rural and outlying districts, notably Jinshan and Chongming, cluster at the lowest end of the pricing spectrum, with median purchase prices dropping below 16,000 Yuan/sqm. These districts exhibit the highest price growth, especially in emerging districts like Qingpu, showing how Shanghai's decentralization is in expansion.

2.3.2. Internal dispersion vs. District averages

A finding derived from the boxplot analysis is that traditional descriptive metrics, such as the arithmetic mean, fail to capture the true distribution of Shanghai's housing market. The averages are pulled upward due to heavy right-skewness caused by luxury properties, hence our resort to the medians and percentiles (especially quartiles). Property outliers identified in the upper fences do not constitute data noise; rather, they reflect genuine, hyper-premium enclaves or ultra-large luxury layouts within central districts.

2.3.3. Property Characteristics and Layout Impact

Beyond geographic location, structural characteristics heavily dictate pricing dynamics, such as layout and size scalability, or the proximity to the metro. Indeed, larger, multi-bedroom layouts naturally correlate with higher aggregate monthly rents and total purchase prices. However, when evaluated on a per-square-metre basis, smaller units (1-bedroom and 2-bedroom studio apartments) frequently command a higher unit premium due to intense demand from young professionals and couples migrating to the city. On the other side, properties tagged with transit-oriented features (such as "近地铁" or near-subway indicators) exhibit a statistically significant price premium and lower market vacancy. This infrastructure premium is particularly acute in Tier 2 transition districts, where access to the metro network serves as the primary gateway for daily commuting into the central business districts.

2.3.4. Investment metrics and Affordability mismatch

By cross-referencing the scraped transactional and rental data with macroeconomic indicators (annual city income and historical price indices), we observe that investors in the center districts are more interested in capital security rather than immediate yields. Central districts operate on low gross rental yields (~1.5% to 1.8%), coupled with high price-to-rent ratios. This implies that buying in the core districts of Shanghai is driven by long-term capital gains as well as security of assets and land scarcity speculation rather than immediate cash-flow generation. The latter could be achieved by the

higher gross rental yields observed in outlying districts, like Jinshan, another type of buyers we evidenced through this data analysis.

When it comes to affordability, the price-to-income analysis highlights a serious strain. For an individual earning the average per-capita income, acquiring a median-priced home in a central district requires an unrealistic multiple of annual earnings, indicating that the urban core is increasingly unattainable without multi-generational wealth or substantial financial leverage.

2.3.5. Data-Schema Implications for Practical Applications

These analytical insights justify the layout of our relational database and final application query parameters. Because a single average doesn't describe appropriately a district, our query script utilizes precise multi-variable filtration which enables users to query exact intersections of district_id, bedrooms, min_area, max_price, and tags LIKE '%近地铁%'. This ensures that the wide internal dispersion and outlier variations observed in our analysis are manageable, providing targeted and realistic property options for the targeted users.

2.4. System Implementation

2.4.1. Database Design

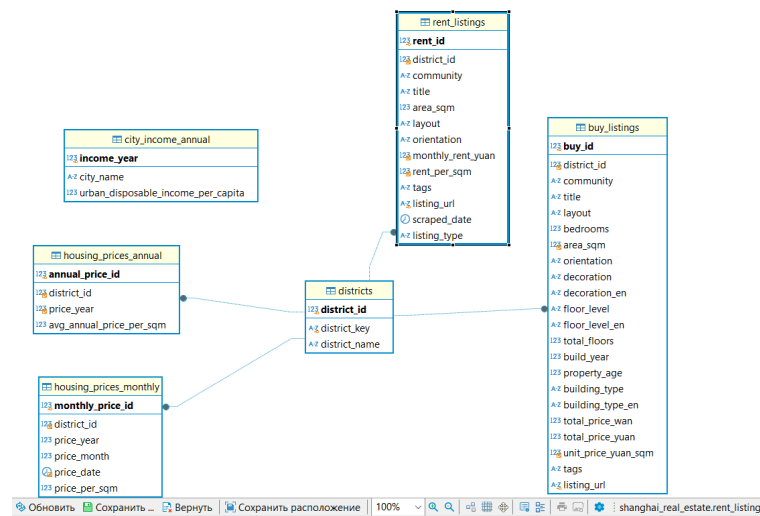
After data collection and preprocessing, the cleaned datasets were stored in a MySQL relational database. The purpose of the database was to organise data from different CSV files, preserve relationships between datasets and allow users to retrieve housing information according to selected conditions.

The database contains six tables:

- districts — Shanghai district reference data;
- buy_listings — current second-hand housing listings;
- rent_listings — current rental listings;
- housing_prices_monthly — monthly historical housing prices;
- housing_prices_annual — annual average housing prices;

- `city_income_annual` — annual disposable income data for Shanghai.

The districts table is used as the main reference table. One district can be connected to many sale listings, rental listings, monthly price records and annual price records. These one-to-many relationships were implemented using the `district_id` foreign key.



Structure and relationships of the MySQL database

The income table is not connected to an individual district because the income dataset represents Shanghai as a whole. Income records can instead be selected according to year.

2.4.2. Data Import into MySQL

Five cleaned CSV files were imported into the database using Python, pandas and PyMySQL.

Source file	MySQL table
lianjia_buy_clean_20260606.csv	buy_listings
lianjia_rent_clean_20260606.csv	rent_listings
shanghai_housing_prices.csv	housing_prices_monthly

shanghai_housing_prices_annual.csv	housing_prices_annual
shanghai_income_final.csv	city_income_annual

The import process first created the district reference records. The program then retrieved the corresponding ***district_id*** for each district and used it when inserting housing listings and historical price data. Parameterised SQL statements with %s placeholders were used to insert values safely. After the import, the number of records in each table was checked using ***SELECT COUNT(*)***.

```
(venv) PS C:\Users\korepanova\Desktop\shhomework3> python .\02_import_csv.py
Buy listings loaded
Rent listings loaded
Monthly prices loaded
Annual prices loaded
Income data loaded

Rows in database:
districts: 16
buy_listings: 2084
rent_listings: 1852
housing_prices_monthly: 2020
housing_prices_annual: 175
city_income_annual: 11
```

2.4.3. Business Query Implementation

A Python interface was created to execute parameterised business queries. Instead of displaying all stored records, users can enter their housing requirements and receive only the matching results.

A. Property Purchase Search

The purchase query allows the user to specify:

- district;
- number of bedrooms;
- minimum property area;
- maximum purchase price.

The database returns the matching community, property size, total price, price per square metre and listing URL.

```

1 - Show all districts
2 - Find sale listings
3 - Find rental listings
4 - Find rental listings by tag
5 - Search by community name
6 - Show annual prices
7 - Show monthly prices
8 - Show city income
9 - Find a listing by URL
0 - Exit
Choose an option: 2
District name: pudong
Bedrooms: 2
Minimum area: 50
Maximum total price: 5000000
Listings found: 40
(809, '金地未来家', '精装修保养好, 高层楼层采光好, 看房方便', 2, Decimal('65.00'), Decimal('3980000.00'), Decimal('61231.00'), 'https://sh.lianjia.com/ershoufang/107115149158.html')
(810, '浦东八村', '店长推荐高性价比, CBD位置, 全明户型', 2, Decimal('61.92'), Decimal('1660000.00'), Decimal('26809.00'), 'https://sh.lianjia.com/ershoufang/107115043122.html')
(812, '高迪北苑', '18年次新盘 高区景观楼层 三开间朝南 诚意出售 有钥匙', 2, Decimal('85.61'), Decimal('3350000.00'), Decimal('39131.00'), 'https://sh.lianjia.com/ershoufang/107115262148.html')
(815, '香山新村北苑', '近地铁, 双南两卫, 原房东急售', 2, Decimal('69.45'), Decimal('2888000.00'), Decimal('41779.00'), 'https://sh.lianjia.com/ershoufang/107115262148.html')
(817, '三林世博家园(东平南苑390弄)', '近地铁6, 11, 13号线, 小区位置好楼层好, 看房方便', 2, Decimal('73.16'), Decimal('3250000.00'), Decimal('44424.00'), 'https://sh.lianjia.com/ershoufang/107115266316.html')
(822, '海康新居', '北三林-海康国际社区+9号线+精装修+好户型', 2, Decimal('97.15'), Decimal('2890000.00'), Decimal('33047.00'), 'https://sh.lianjia.com/ershoufang/107115294673.html')
(829, '金海新城御景苑', '双地铁 德忌利士 南苑上车 南北通 采光视野好', 2, Decimal('79.54'), Decimal('2290000.00'), Decimal('28791.00'), 'https://sh.lianjia.com/ershoufang/107115336623.html')
(829, '恒大金地未来家(二期)', '北高南 南北两居一厅4-8-11三轨相交, 采光好 楼层好', 2, Decimal('68.90'), Decimal('3450000.00'), Decimal('50073.00'), 'https://sh.lianjia.com/ershoufang/107115180649.html')
(830, '妙境一村', '近地铁, 出行便利, 户型方正, 南北格局通大两房', 2, Decimal('80.54'), Decimal('2180000.00'), Decimal('27068.00'), 'https://sh.lianjia.com/ershoufang/107115233558.html')
(837, '恒耀东郡', '山阳多付可送, 地铁700米, 南北格局采光通风佳', 2, Decimal('89.40'), Decimal('3750000.00'), Decimal('41947.00'), 'https://sh.lianjia.com/ershoufang/107115273827.html')
(838, '万泰家园', '北高南 南北通 全明户型 采光通风好 中间楼层 位置', 2, Decimal('77.00'), Decimal('2780000.00'), Decimal('36104.00'), 'https://sh.lianjia.com/ershoufang/107115229897.html')
(840, '海康国际', '近地铁, 开发商精装修 少有高楼层 采光充足, 真拎包入住', 2, Decimal('77.71'), Decimal('1890000.00'), Decimal('24322.00'), 'https://sh.lianjia.com/ershoufang/107115341220.html')

```

Property purchase search based on user requirements.

This query supports prospective buyers by reducing a large dataset to properties that satisfy their main requirements.

B. Rental Property Near a Metro Station

The rental query allows the user to specify:

- district;
- minimum area;
- maximum monthly rent;
- metro proximity requirement.

The metro condition is checked using the LIKE operator and the 近地铁 tag contained in the rental listings.

```

Only the first 20 rows are shown

1 - Show all districts
2 - Find sale listings
3 - Find rental listings
4 - Find rental listings near metro
5 - Search by community name
6 - Show annual prices
7 - Show monthly prices
8 - Show city income
9 - Find a listing by URL
0 - Exit

Choose an option: 4
District name: pudong
Minimum area: 10
Maximum monthly rent: 50000
Rows found: 52
(803, '金杨新村三街坊', '整租·金杨新村三街坊 1室1厅 南', Decimal('46.19'), '1室1厅1卫', Decimal('4305.00'), Decimal('93.20'), '自售|新上|月租|近地铁|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH181409595415313252.html')
(804, '30.00m', '独栋·南城公寓 圣江店【00后看这里】2号线地铁口, 精装修, 免费用水免费上网 1室1厅', Decimal('30.00'), '1室1厅1卫', Decimal('3400.00'), Decimal('113.30'), '独栋公寓|月租|拎包入住|近地铁|开敞厨房', 'https://sh.lianjia.com/apartment/107113.html')
(807, '锦河苑', '整租·锦河苑 2室2厅 南', Decimal('90.59'), '2室2厅1卫', Decimal('6500.00'), Decimal('71.80'), '官方核验|新上|近地铁', 'https://sh.lianjia.com/zufang/SH2172936035955113984.html')
(808, '高科西路2763弄', '整租·高科西路2763弄 1室1厅 南/北', Decimal('43.42'), '1室1厅1卫', Decimal('4600.00'), Decimal('105.90'), '官方核验|自售|新上|月租|近地铁|精装修|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH2522359133462863072.html')
(810, '齐河花苑', '整租·齐河花苑 2室1厅 南', Decimal('64.29'), '2室1厅1卫', Decimal('4990.00'), Decimal('77.60'), '自售|新上|月租|近地铁|精装修|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH1660278176154124288.html')
(812, '金都新城东区', '整租·金都新城东区 1室1厅 南', Decimal('49.27'), '1室1厅1卫', Decimal('4095.00'), Decimal('86.60'), '自售|新上|月租|近地铁|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH1676852217874743296.html')
(813, '仅卿1间', '独栋·老翠华庭青萃社区 康新公路店【地铁口千源分寓】康桥地铁口, 阳光一居, 带独立卫浴, 带阳台和外窗 1室1厅', Decimal('43.00'), '1室1厅1卫', Decimal('2810.00'), Decimal('65.38'), '独栋公寓|月租|近地铁|精装修|带阳台|开敞厨房|押一付一', 'https://sh.lianjia.com/apartment/96107.html')
(814, '锦河苑东区', '整租·锦河苑东区 2室1厅 南', Decimal('55.07'), '2室1厅2卫', Decimal('5000.00'), Decimal('90.10'), '自售|新上|近地铁|精装修|押一付一|双卫生间|随时看房', 'https://sh.lianjia.com/zufang/SH172897926401801356.html')
(815, '东方大景苑', '整租·东方大景苑 3室2厅 南', Decimal('143.67'), '3室2厅2卫', Decimal('14000.00'), Decimal('97.40'), '官方核验|业主自荐|新上|近地铁|精装修|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH1773632855238721536.html')
(816, '都汇公馆', '整租·都汇公馆 1室1厅 东/东南', Decimal('51.93'), '1室1厅1卫', Decimal('2100.00'), Decimal('40.40'), '自售|新上|近地铁|精装修|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH1758744653379235840.html')
(818, '梅园六街坊', '整租·梅园六街坊 2室1厅 南', Decimal('52.78'), '2室1厅1卫', Decimal('6930.00'), Decimal('133.80'), '自售|新上|近地铁|精装修|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH1773632855238721536.html')
(819, '梅园新村(浦东)', '整租·梅园新村(浦东) 2室1厅 南', Decimal('54.00'), '2室1厅1卫', Decimal('4200.00'), Decimal('77.80'), '自售|新上|近地铁|押一付一|随时看房', 'https://sh.lianjia.com/zufang/SH1776547650657583104.html')

```

Rental listings within the selected budget and marked as being near a metro station.

This query combines numerical filtering with text-based search. It allows users to find rental properties that meet their area and budget requirements and are described on Lianjia as being close to a metro station. The query does not calculate the exact distance to a station; it uses the property tag provided in the original listing.

C. Historical Housing Price Search

The historical price query allows the user to specify:

- district;
- year.

The database returns monthly housing prices per square metre for the selected district and year. This query can be used to examine price

changes over time and provide data for further visualisation.

```

1 - Show all districts
2 - Find sale listings
3 - Find rental listings
4 - Find rental listings near metro
5 - Search by community name
6 - Show annual prices
7 - Show monthly prices
8 - Show city income
9 - Find a listing by URL
0 - Exit
Choose an option: 7
District name: pudong
Year: 2025
Rows found: 12
(2025, 1, datetime.date(2025, 1, 1), Decimal('69128.00'))
(2025, 2, datetime.date(2025, 2, 1), Decimal('68141.00'))
(2025, 3, datetime.date(2025, 3, 1), Decimal('67300.00'))
(2025, 4, datetime.date(2025, 4, 1), Decimal('69200.00'))
(2025, 5, datetime.date(2025, 5, 1), Decimal('65000.00'))
(2025, 6, datetime.date(2025, 6, 1), Decimal('64900.00'))
(2025, 7, datetime.date(2025, 7, 1), Decimal('63800.00'))
(2025, 8, datetime.date(2025, 8, 1), Decimal('64000.00'))
(2025, 9, datetime.date(2025, 9, 1), Decimal('66400.00'))
(2025, 10, datetime.date(2025, 10, 1), Decimal('66000.00'))
(2025, 11, datetime.date(2025, 11, 1), Decimal('66000.00'))
(2025, 12, datetime.date(2025, 12, 1), Decimal('66400.00'))

```

Monthly housing prices retrieved from the MySQL database

The query returns the year, month, date and housing price per square metre. These records can be loaded into a pandas DataFrame and used to create a line chart.

3. RESULTS AND FINDINGS

3.1. Conclusion

This project successfully designed and implemented a functional web scraping pipeline and a subsequent database system to improve public transparency regarding real estate affordability in Shanghai. On the technical front, we overcame data collection obstacles, such as website anti-scraping mechanisms and dynamic listing layout, by configuring Selenium (as seen in class) to automate browser interactions across multiple product formats. The scraped raw text files were cleaned, processed, and structured into a relational MySQL database containing six distinct tables to preserve key entity relationships.

To make this data actionable for public use, we built a terminal-based interface using Python and PyMySQL. This command-line tool (CLI) executes business queries that drastically reduce complex data into filtered user requirements, such as multi-variable evaluations matching specific district IDs, bedroom counts, price caps, and transit proximity keywords. Ultimately, the technical part of this study successfully demonstrates how data collection tools, automated data cleaning, and relational database systems can be integrated to increase public access to market information and support evidence-based decisions for prospective homebuyers and renters.

3.2. Insights

- Spatial price disparities

We evidenced a clear polarization between Shanghai Inner ring and the Middle & Outer rings, establishing distinct premium tiers, with central districts showcasing major pricing advantages.

- Market softening trends

When comparing current listing medians with historical data, we revealed a general decline below 2025 annual averages across most districts, which could be due to market softening or to shifts in listing composition.

- Distribution distorted by luxury outliers

The purchase and rental price distributions are “right-skewed” meaning that the means are pulled upward by ultra-premium luxury properties, we often suggested to remove the top 1% for this reason.

- Rental yield mismatch

Central business zones show high property values but low gross rental yields, as opposed to outlying districts, more prone to purely income-focused investments.

- Affordability strain

Comparing current property price trends against income indices show an important financial burden for buyers relying on average incomes, which makes the acquisition of a home in the Inner ring almost impossible.

3.3. Limitations and Future Improvements

While this project provides valuable data-driven insights into the Shanghai real estate market, several critical limitations should be addressed in future work, notably the following.

- Data scope and representation bias

The dataset relies entirely on listings scraped from a single online real estate platform (Lianjia) sampled during a specific window in June 2026, with a limit of 5 results pages scraped per district (for technical reasons, due to limitations...). Future iterations should dynamically scrape multiple real estate platforms over a multi-month period to build a more comprehensive panel dataset, ensuring more precision in the insights.

- Granularity and outliers limitations

The boxplot analysis revealed heavy internal price dispersion and heavy right-skewed layout outliers within several central districts, proving that broad district-level aggregates can mask domestic luxury enclaves or unusual property types. The current analysis lacks spatial granularity, such as specific sub-districts, neighborhoods, or geolocated coordinates (as an order of magnitude, Minhang’s population was 2,65 million in 2020). Incorporating GIS mapping and sub-district boundaries would dramatically improve geographic precision.

- Variables and modelling simplifications

Notably, important non-spatial property characteristics, such as precise proximity to local metro stations (currently relying on simple text-based tags like "近地铁" rather than calculated walking or bike distances), architectural building styles, specific decoration or service levels, and proximity to top-tier schools or hospitals, were either simplified or excluded due to incomplete web data fields on Lianjia.

- Technical CLI interactivity (vs. a dashboard)

Although the terminal-based Python business query program effectively handles detailed perimeter-based SQL queries for consumers, it lacks intuitive visualization and scalability for non-technical users. Future technical enhancements should migrate this backend SQL schema to an interactive, web-based dashboard interface (e.g., using Streamlit, which was mentioned in the Syllabus) featuring real-time geographic heatmaps and interactive filters to better advise prospective tenants and buyers.

3.4. Task Division

Name	Cha Jie Min	Addaleia Jaedden C. Williamson	Anastasiia Korepanova	Jean-Baptiste Mathieu
Student ID	524120990049	525082990002	723120290022	723370290010
Data Collection	✓	✓		✓
Data Preprocessing	✓	✓		
Data Analysis			✓	✓
Database Implementation			✓	
Visualization		✓	✓	

Report Writing	✓	✓	✓	✓
Presentation	✓	✓	✓	✓

3.5. Project Repository

The complete source code, datasets, database implementation scripts, and project documentation can be accessed through the project's GitHub repository:

[BUSS2510-01 Course Project](#)

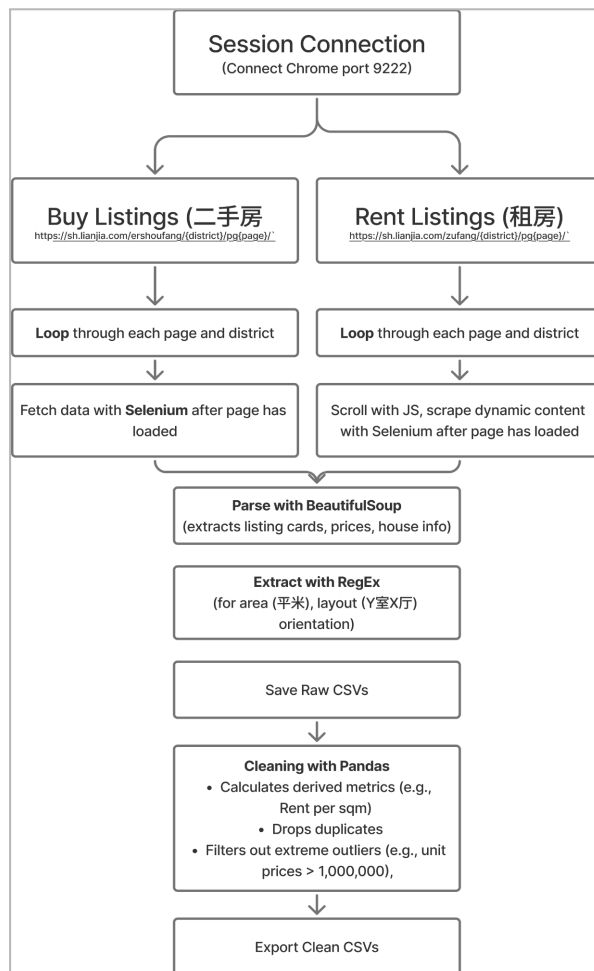
The repository contains the web scraping programs, data preprocessing workflows, MySQL database implementation, business query interface, and visualization code used throughout this study.

REFERENCES

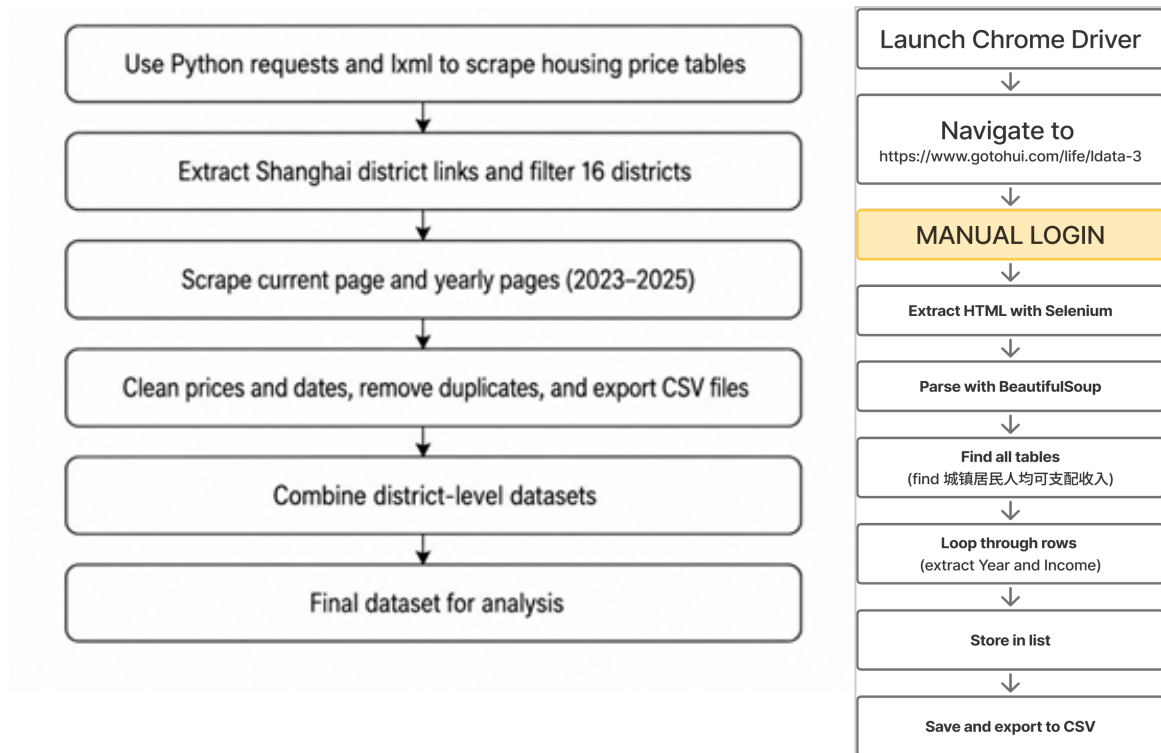
- Chen, J., Chen, Y., Hill, R. J., & Hu, P. (2022). The user cost of housing and the price-rent ratio in Shanghai. *Regional Science and Urban Economics*, 92, Article 103738. <https://doi.org/10.1016/j.regsciurbeco.2021.103738>
- Han, L., Wei, Y. D., & Wu, Y. (2019). Analyzing the private rental housing market in Shanghai with open data. *Land Use Policy*, 85, 271–284. <https://doi.org/10.1016/j.landusepol.2019.04.004>
- Wu, J., Gyourko, J., & Deng, Y. (2020). Evaluating the risk of Chinese housing markets: What we know and what we need to know. *China Economic Review*, 39, 91-114. <https://doi.org/10.1016/j.chieco.2016.03.008>

APPENDIX

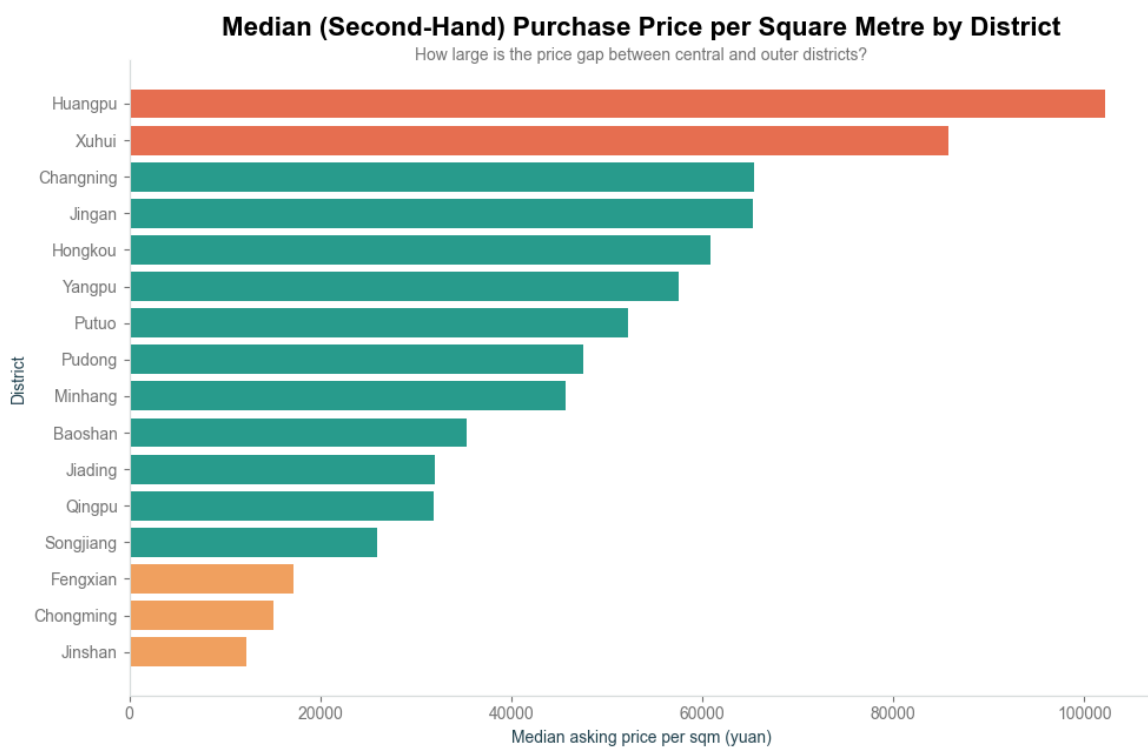
A. Data Collection



<pre> <div class="kem_house-title-ershou" data-id="..."> <div class="house-text"> <div class="house-desc">4室2厅/183.01m²/南北/绿城上海御园</div> </div> <div class="house-tags"> 地铁 VR房源 满五年 有电梯 </div> <p class="house-price"> 1599万 87,373元/平 </p> </div> </pre>	<pre> <div class="content_list-item" data-house_code="SH2034553316041555968"> <div class="content_list-item-main"> <!-- Title --> <p class="content_list-item-title"> 整租:瑞和明庭南区 2室2厅 南 </p> <!-- Description (Location) --> <p class="content_list-item-des"> 青浦 徐汇 瑞和明庭南区 </p> </div> <p class="content_list-item-bottom online"> <div class="content_item_tag-ziying_sxz">自置</div> <div class="content_item_tag-deposit_lpay_1">押一付一</div> <div class="content_item_tag-ls_key">随时看房</div> </p> <p class="content_list-item-brand online"> 贝壳优选 </p> <p class="content_list-item-price"> 3675 元/月 </p> </div> </pre>
Second-Hand Housing <i>Listing Card Structure</i>	Rental Housing <i>Listing Card Structure</i>

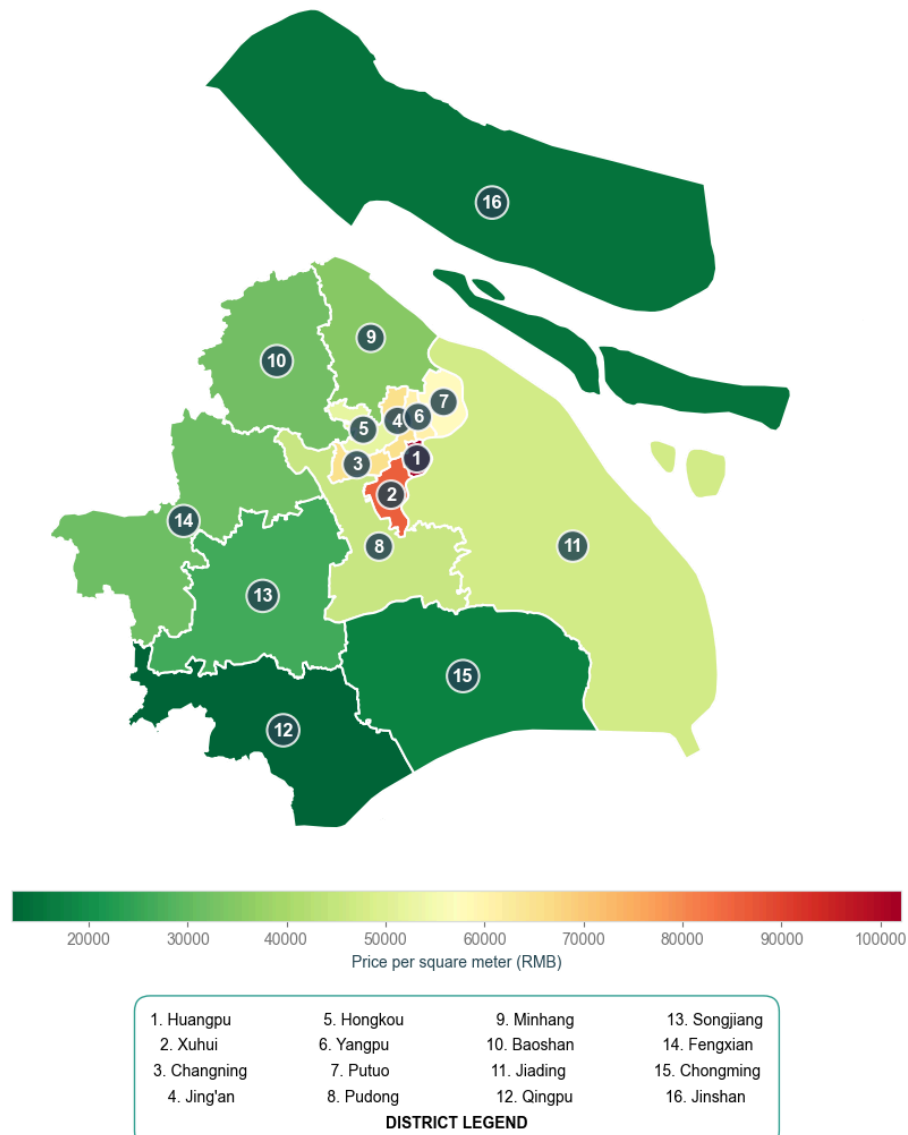


B. Data Visualizations



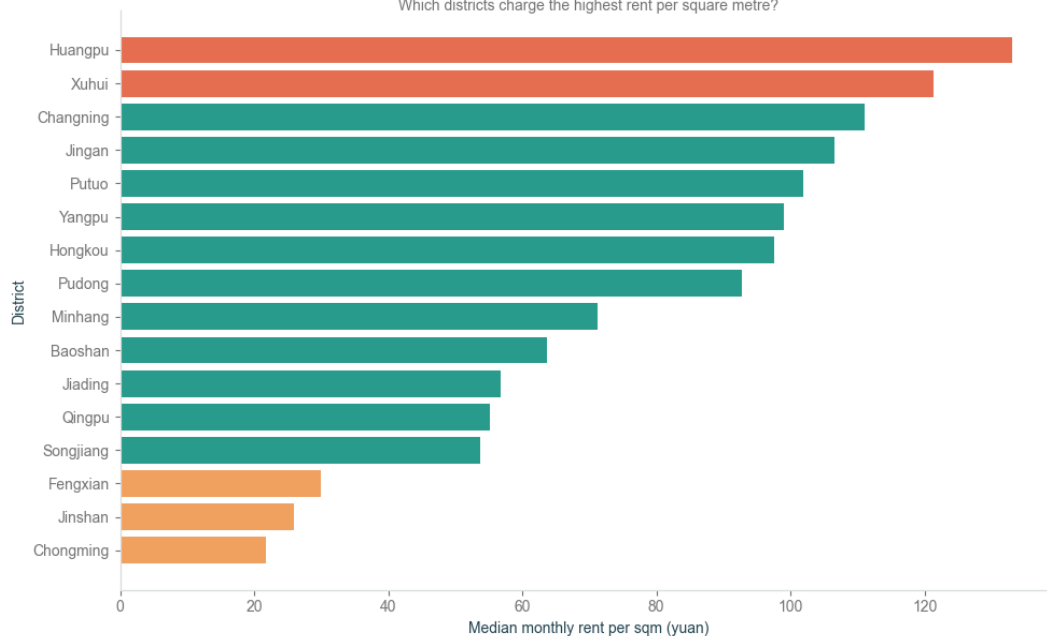
Median Second-Hand Property Prices Across Shanghai June 2026

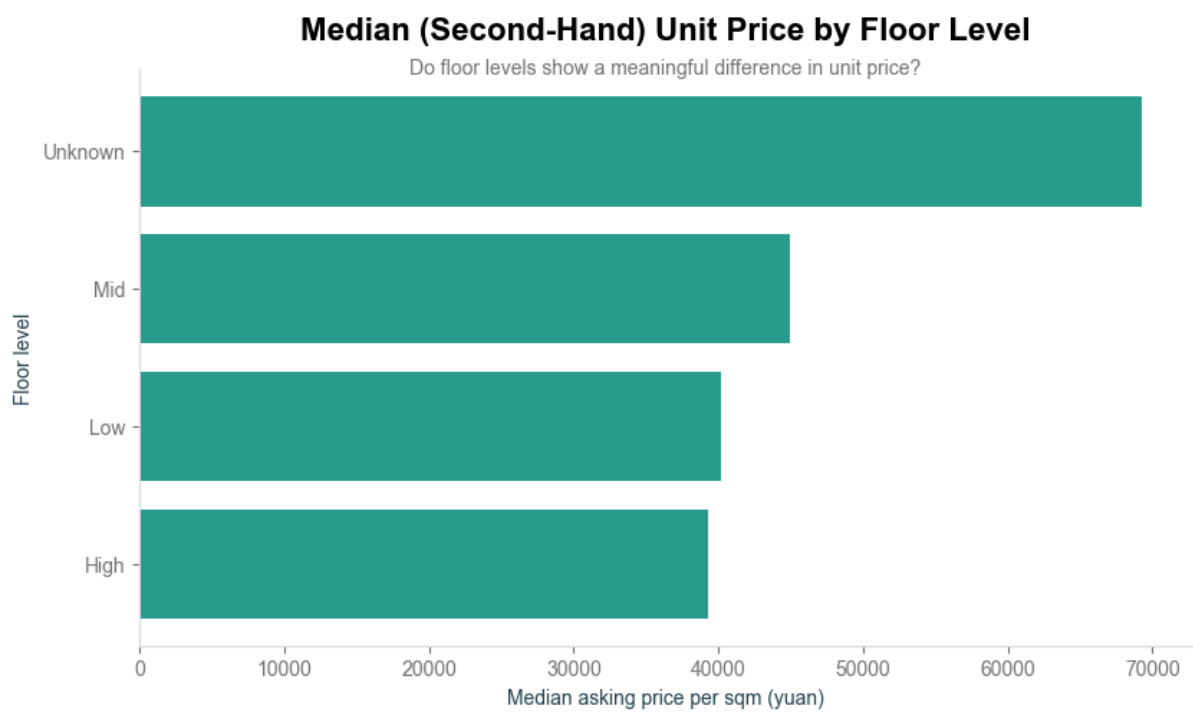
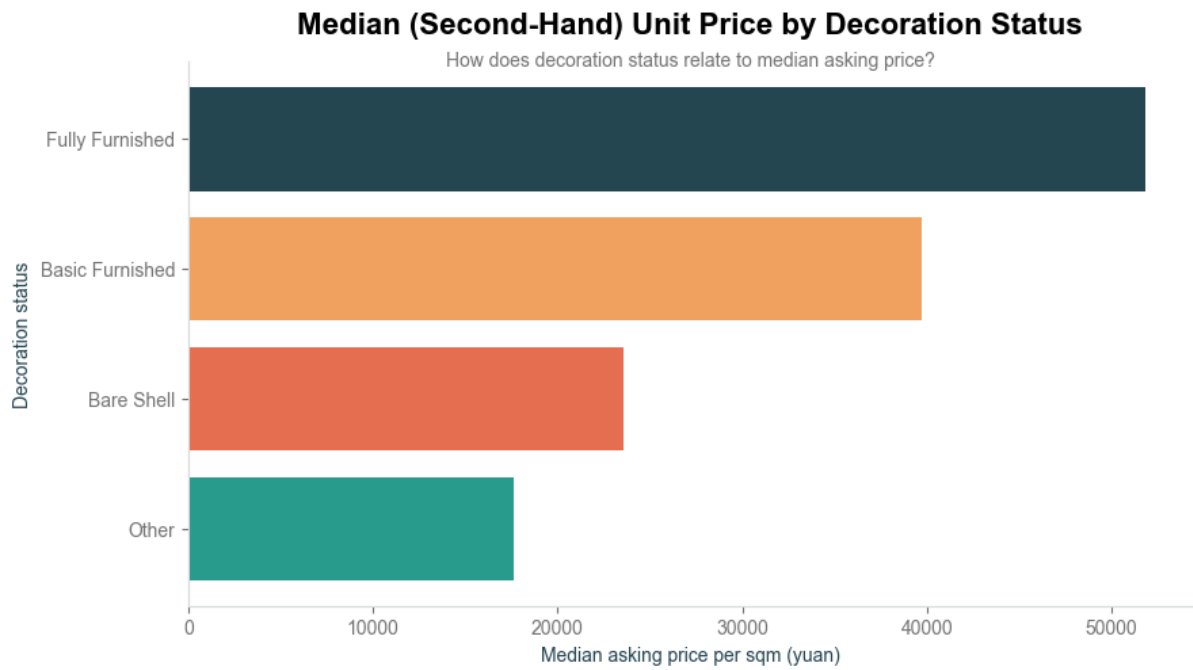
Which districts have the highest property prices?



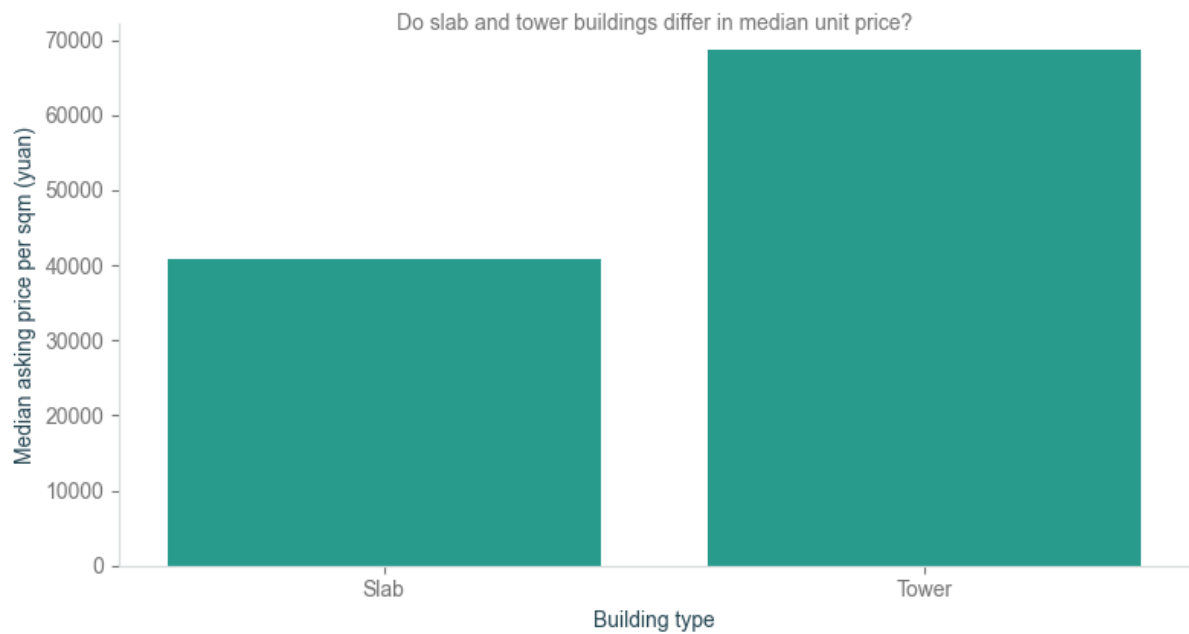
Median Rent per Square Metre by District

Which districts charge the highest rent per square metre?

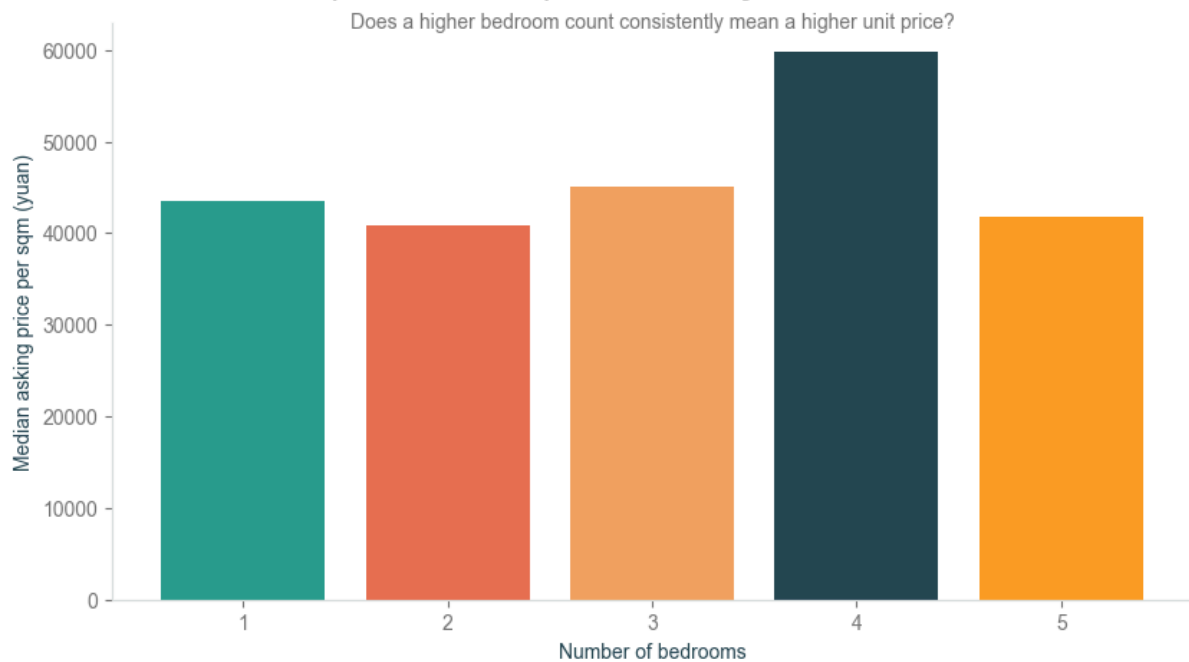


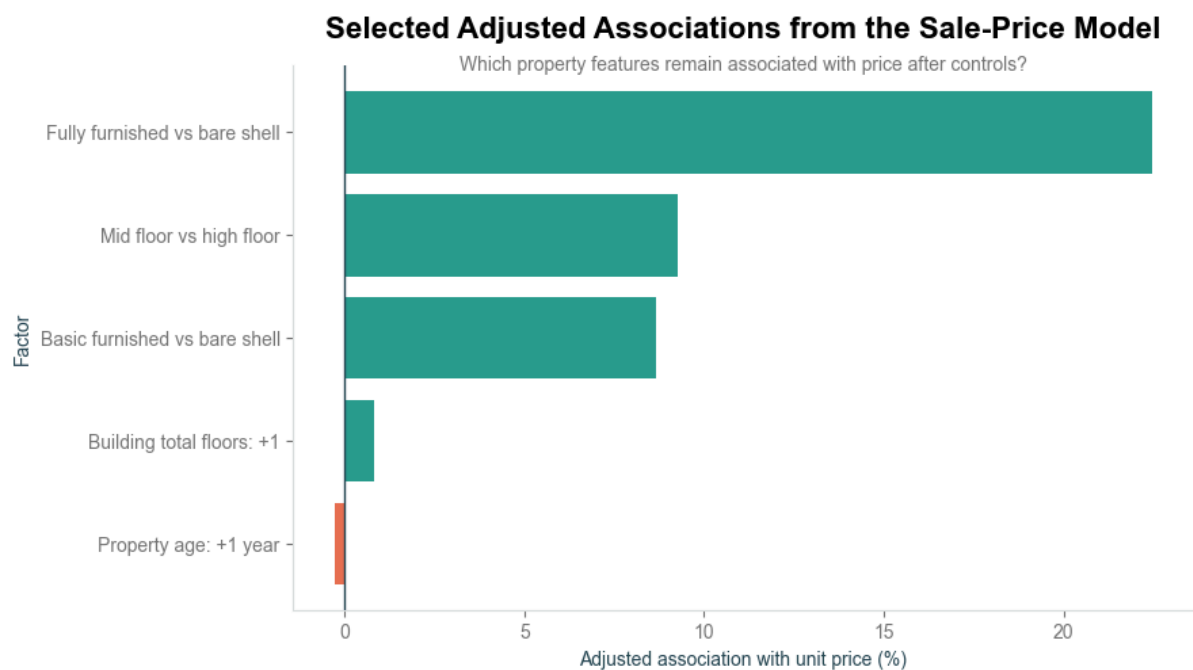
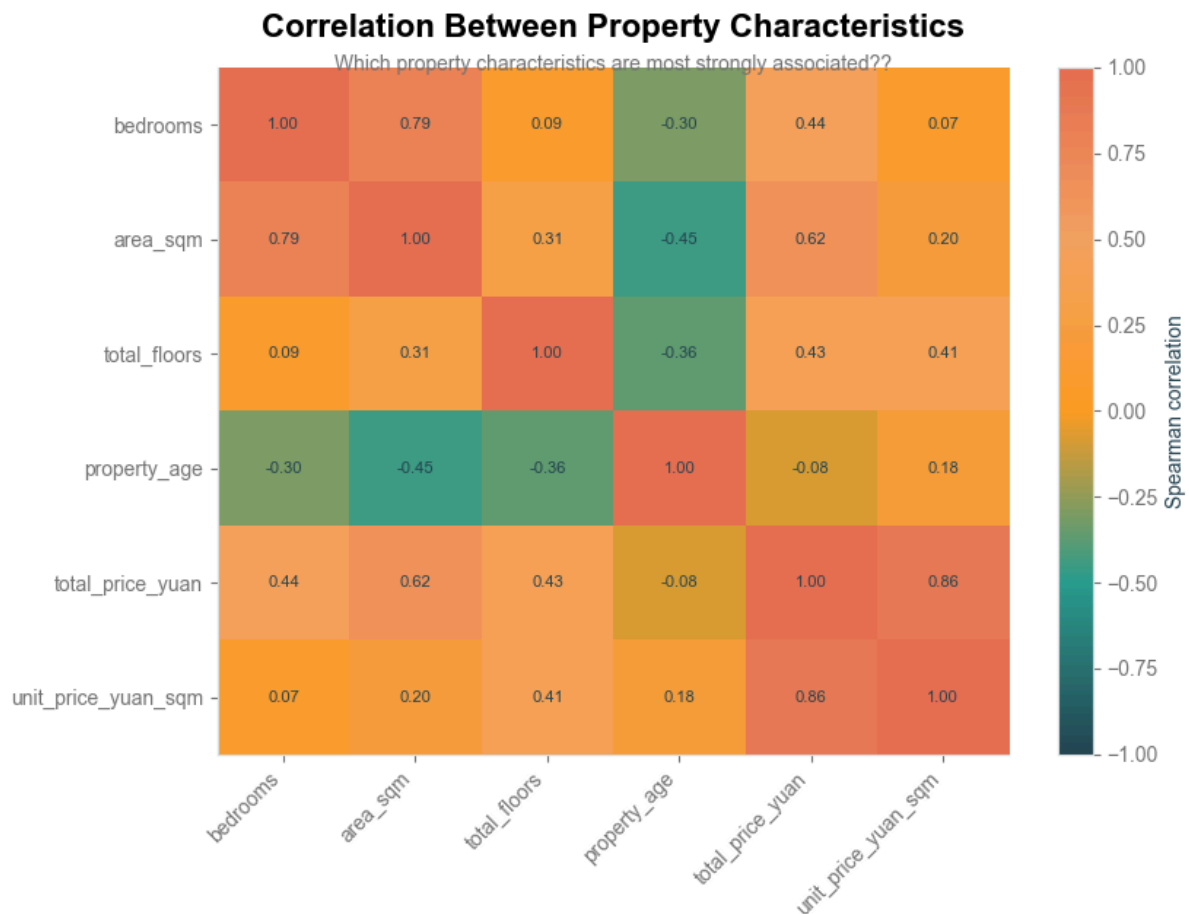


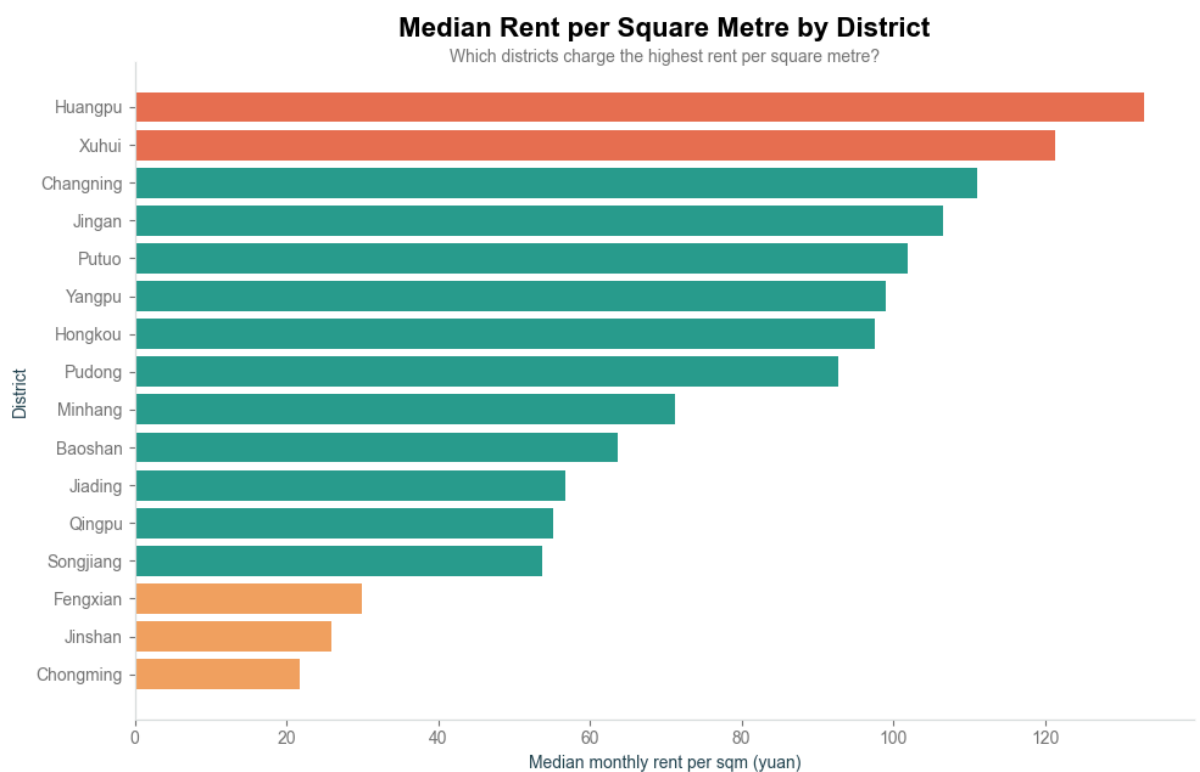
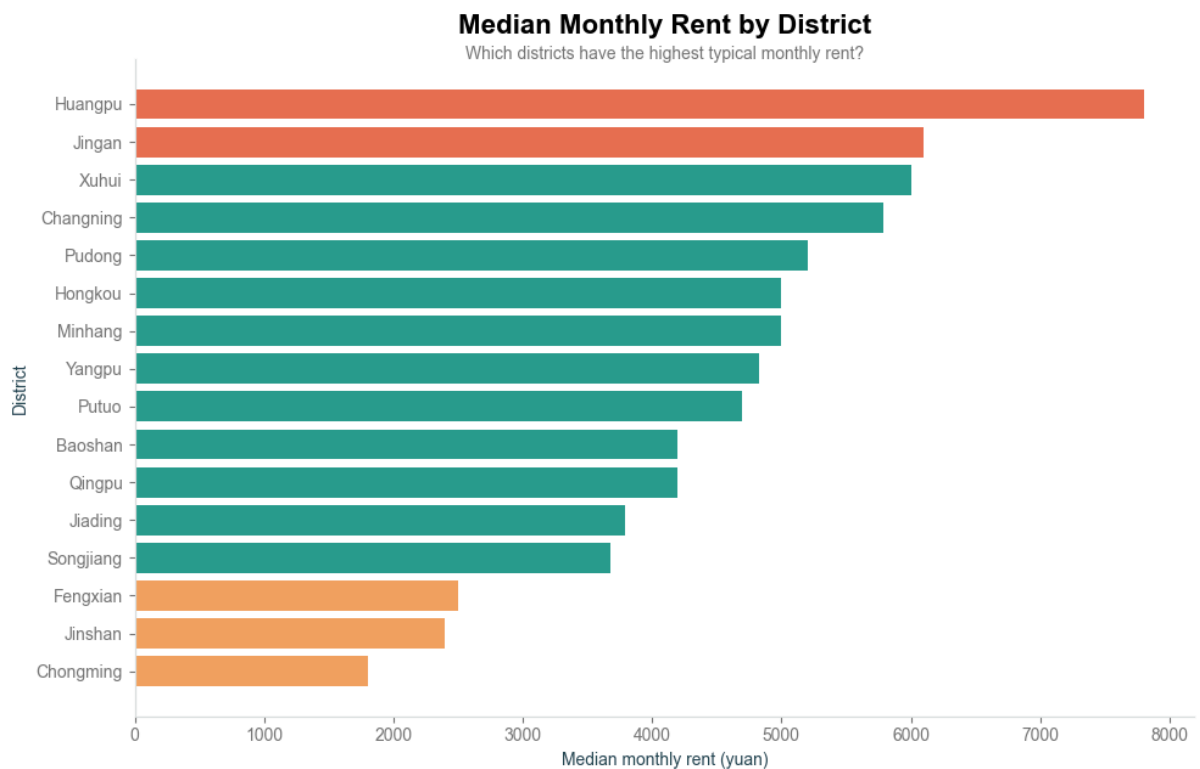
Median (Second-Hand) Unit Price: Slab versus Tower Buildings

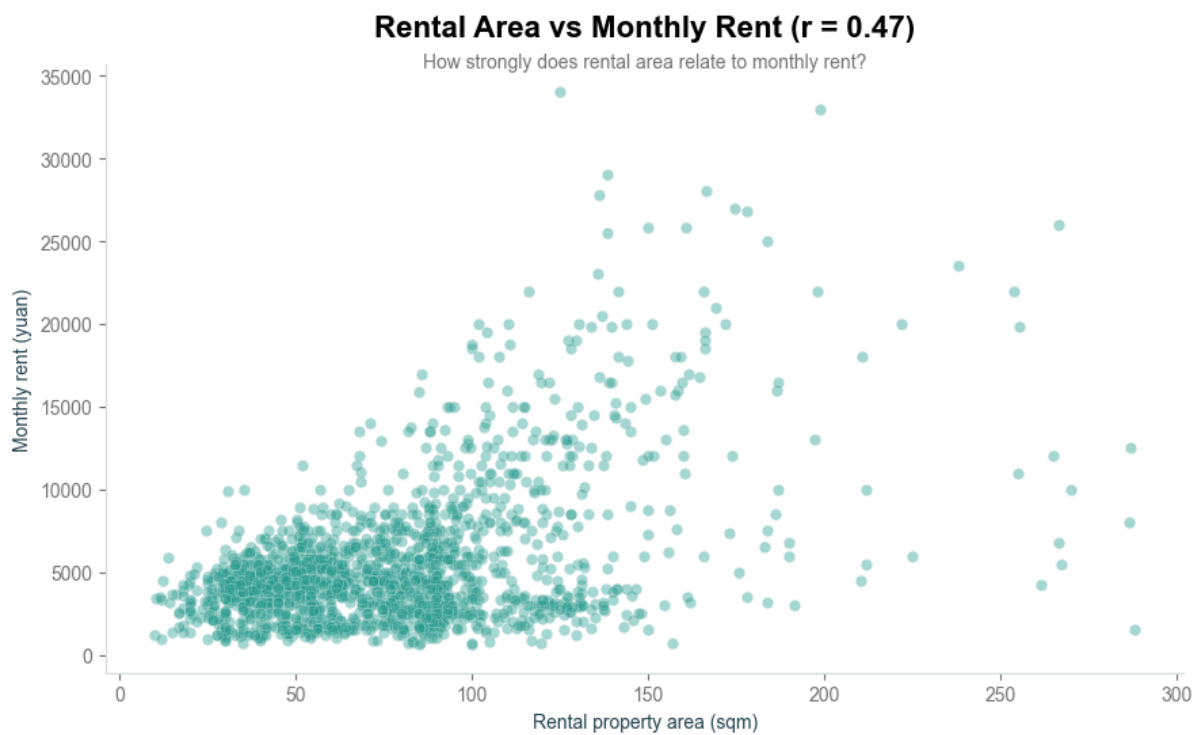
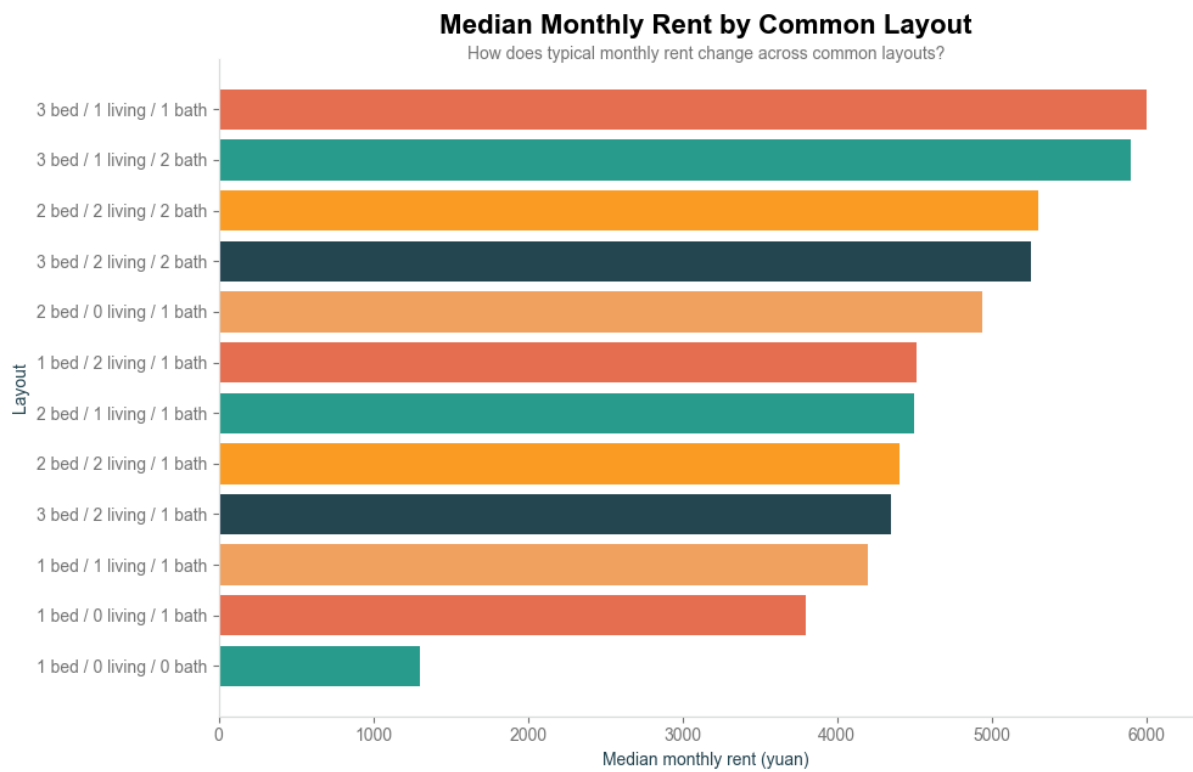


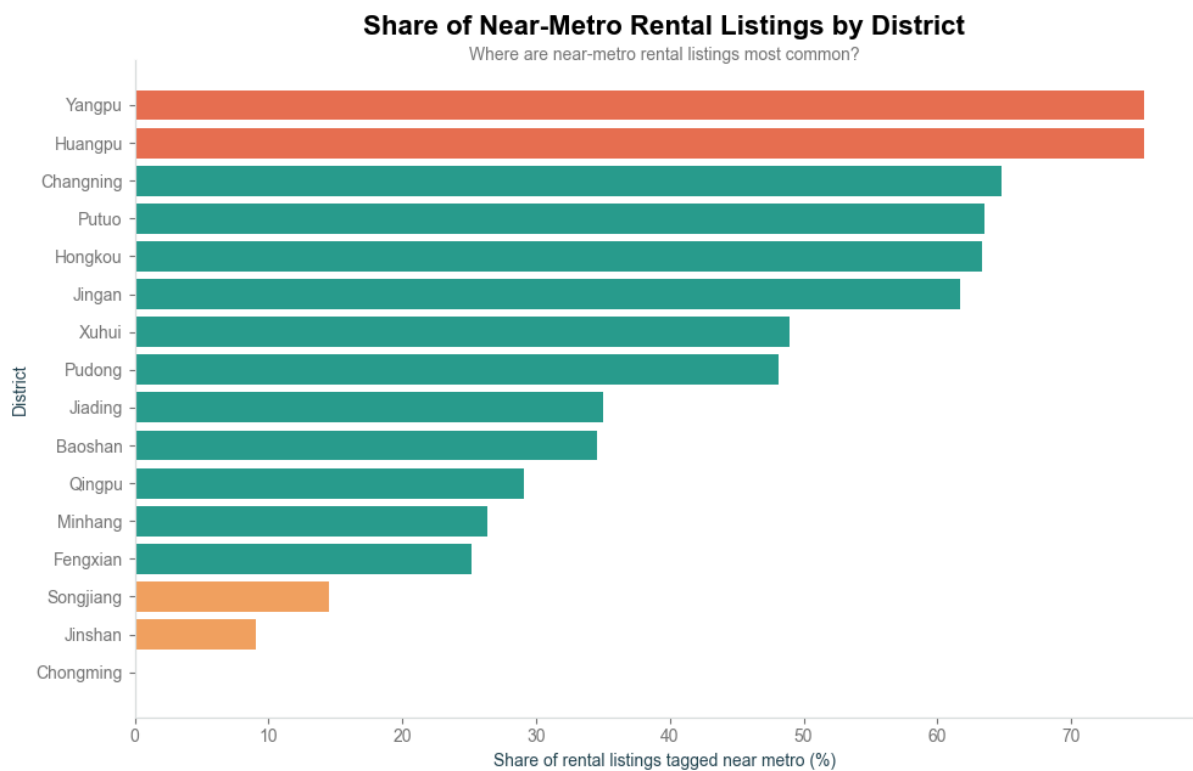
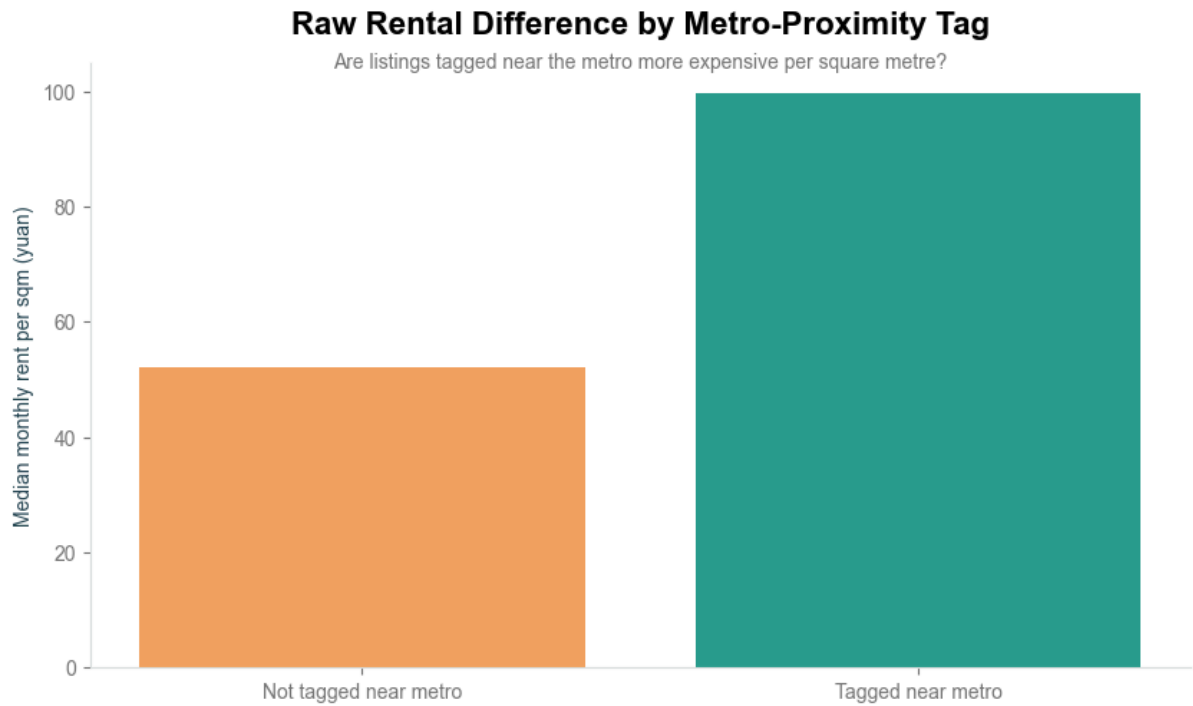
Median (Second-Hand) Unit Price by Number of Bedrooms

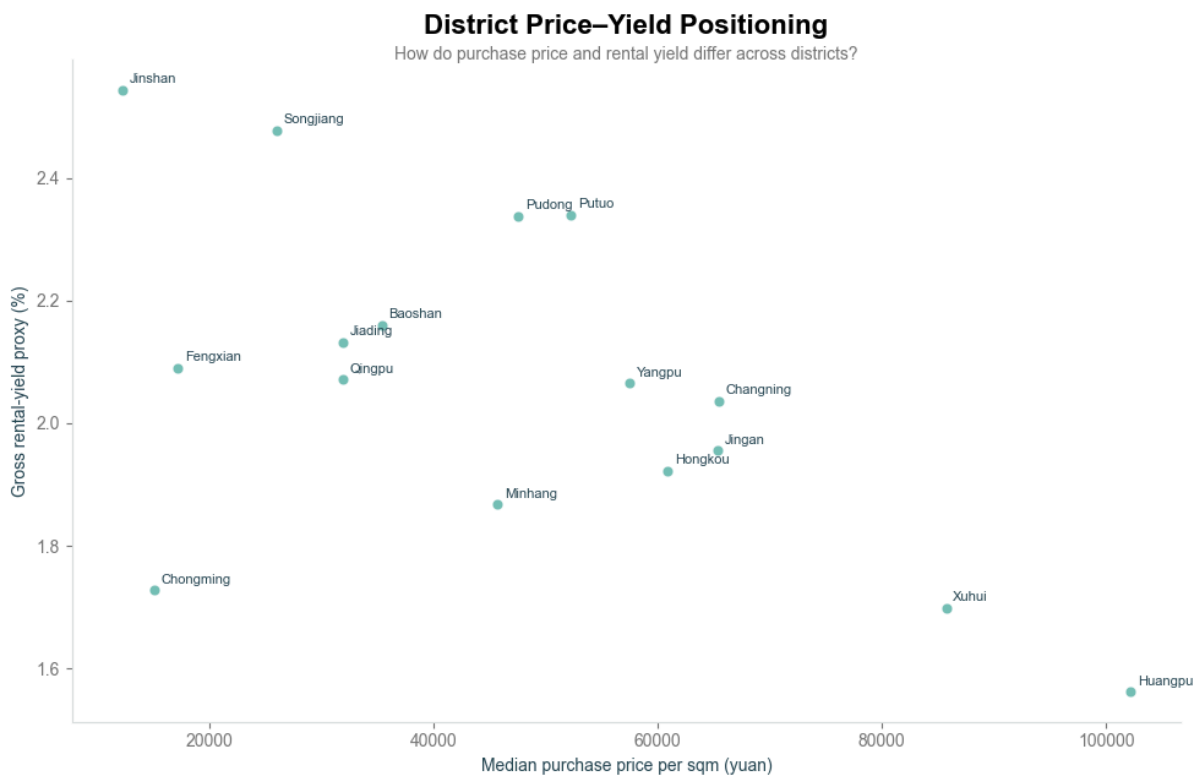
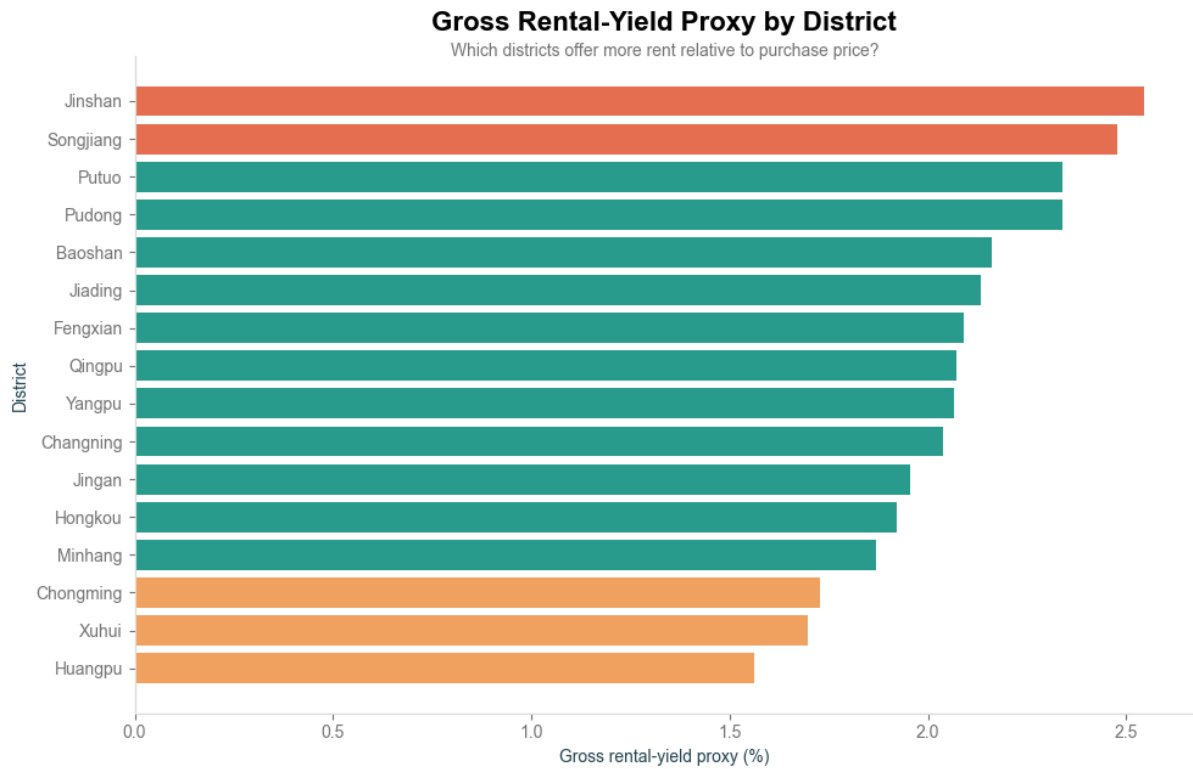






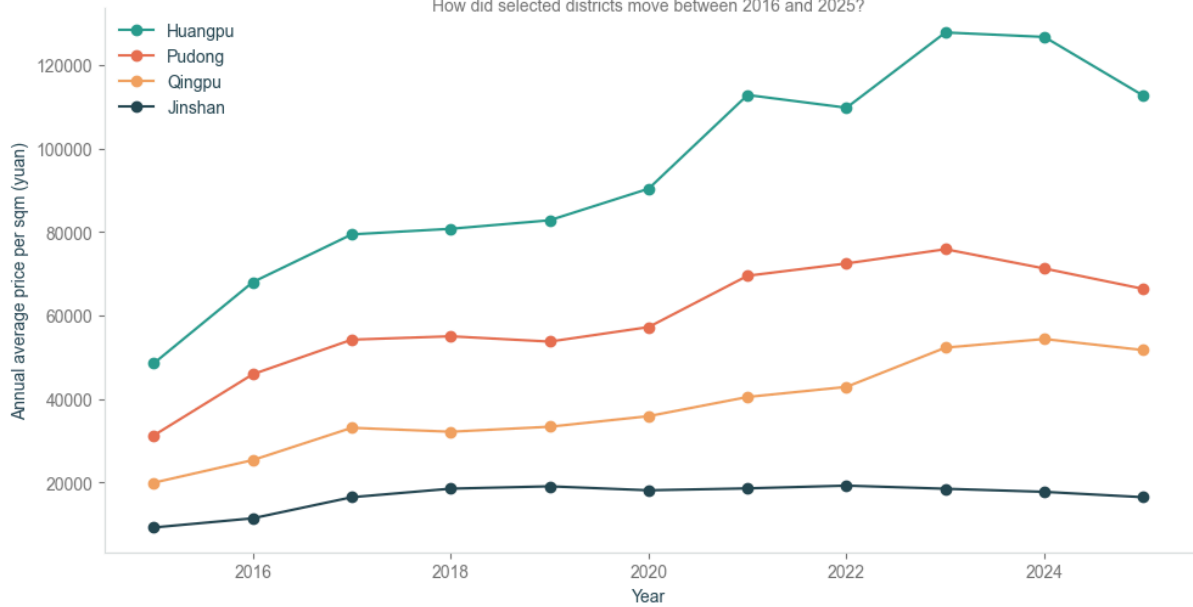






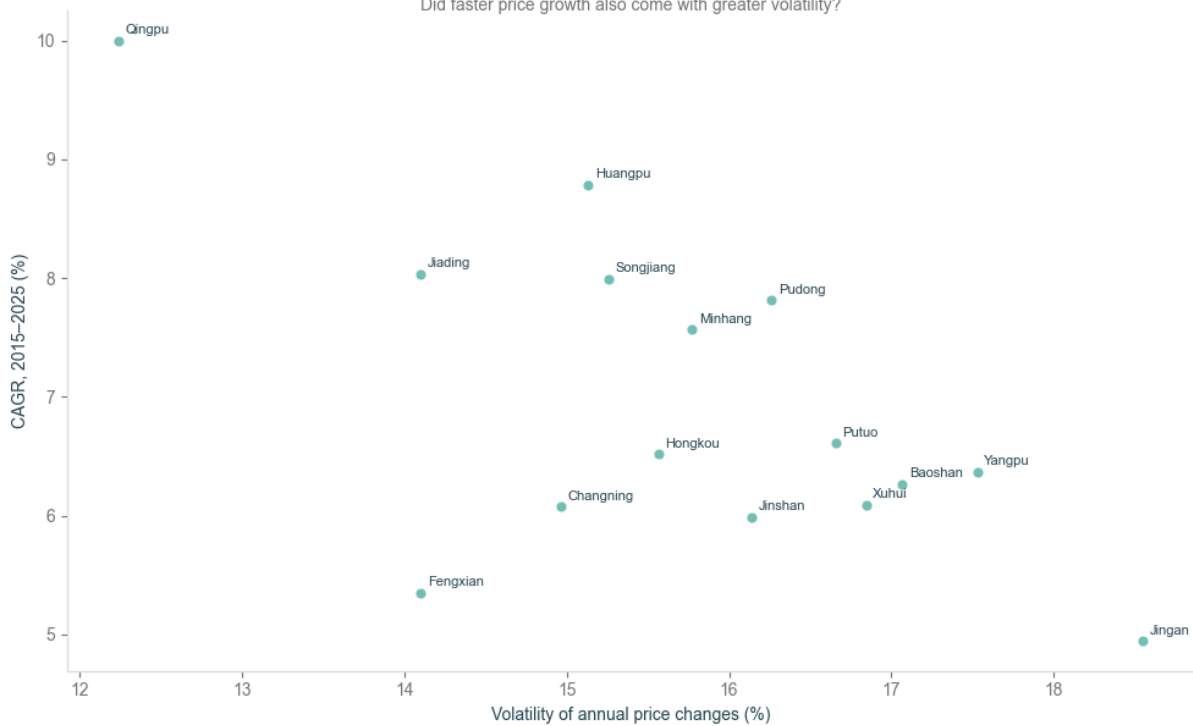
Historical Price Trends for Selected Districts

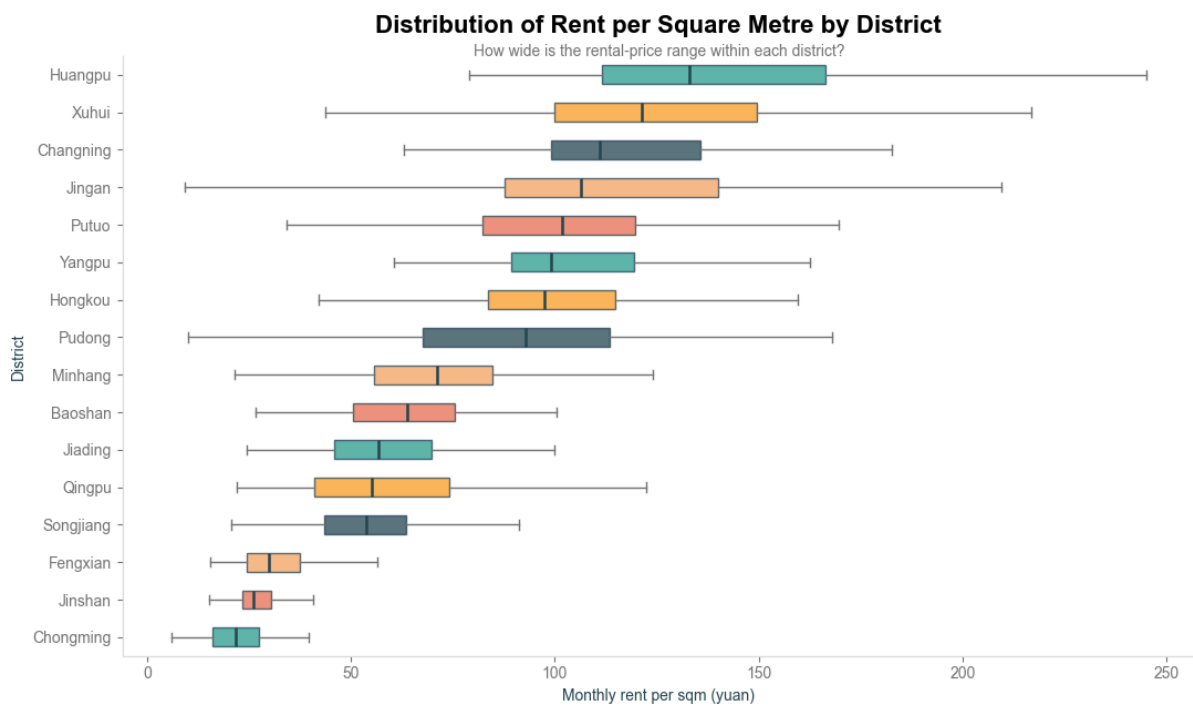
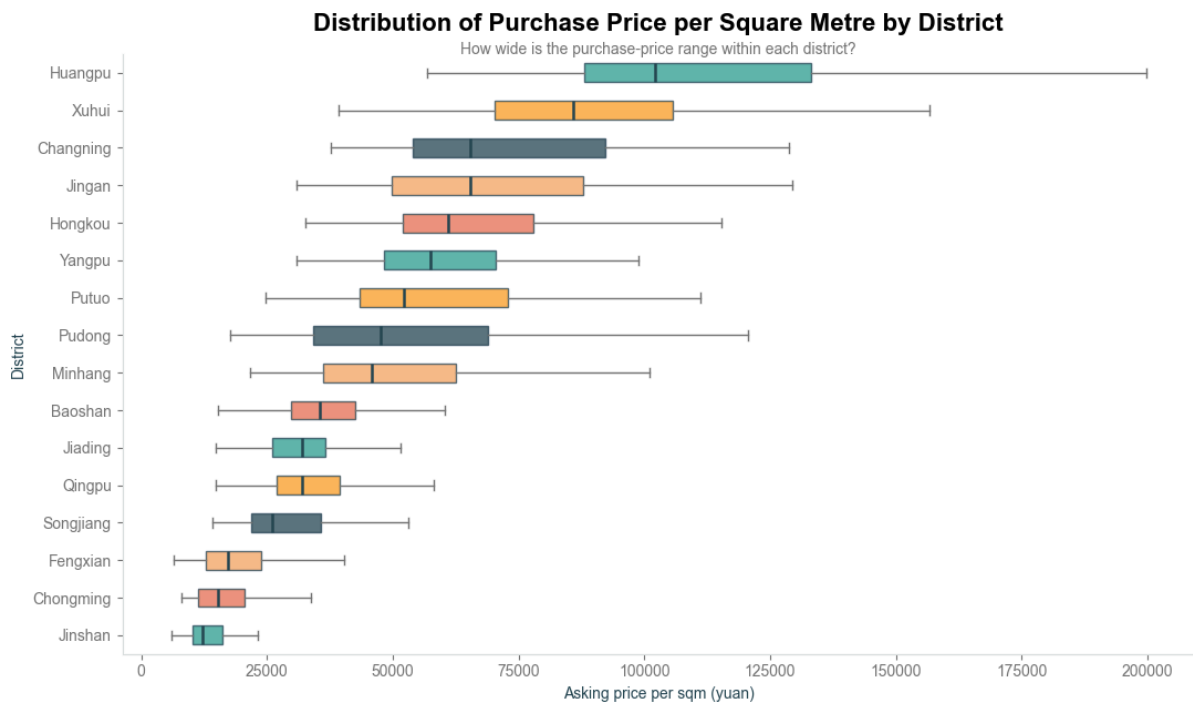
How did selected districts move between 2016 and 2025?



Historical Growth versus Volatility

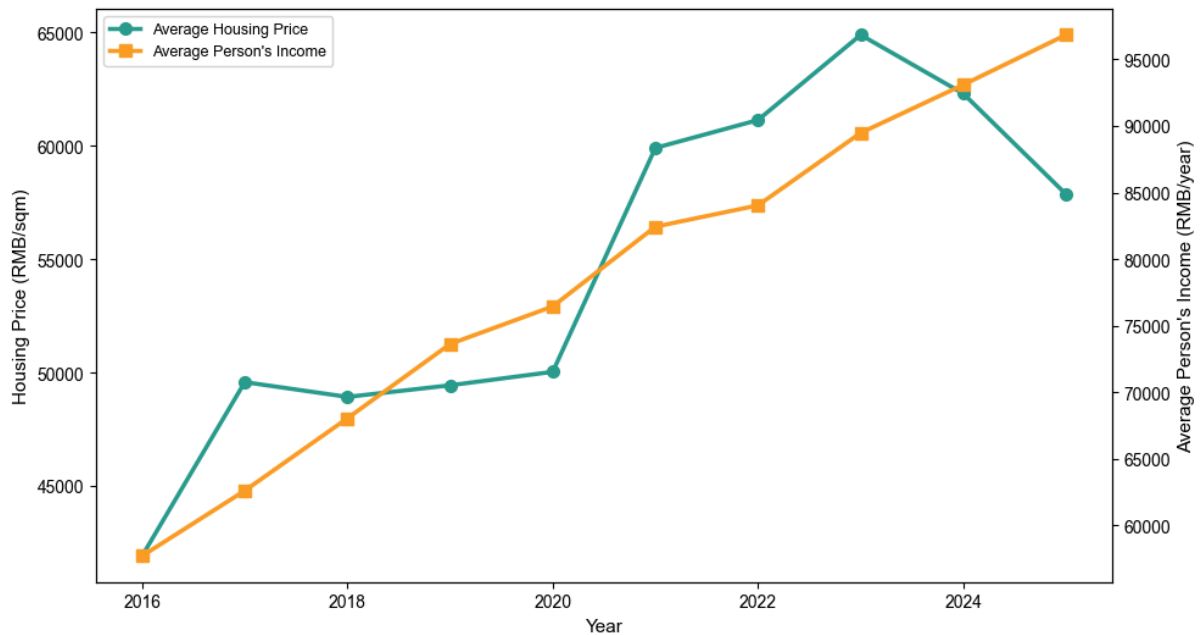
Did faster price growth also come with greater volatility?





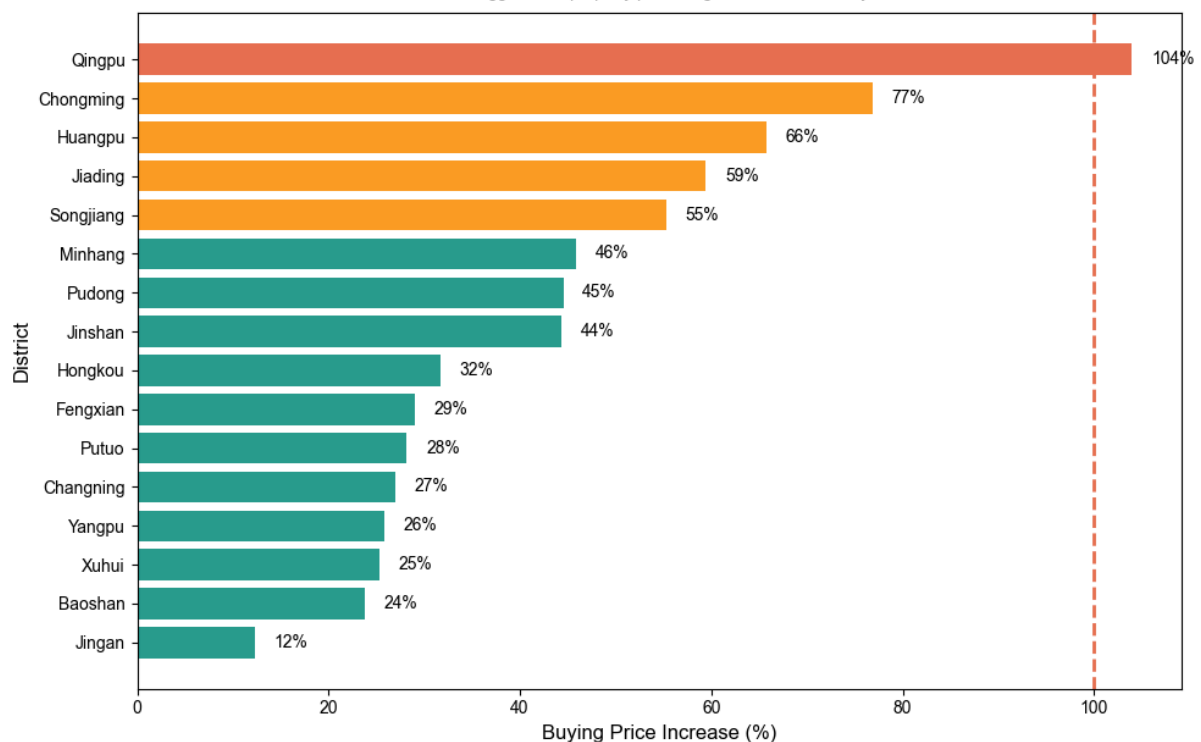
Housing Prices vs Take-home Income in Shanghai 2016 → 2025

Has the average person's take-home pay kept up with rising housing prices?



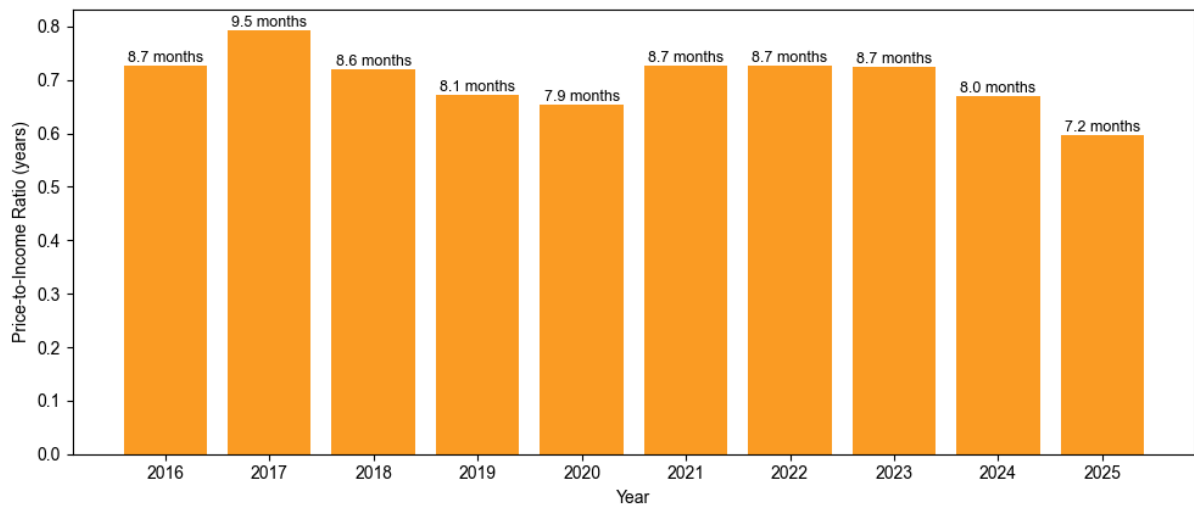
Shanghai Housing Price Growth by District 2016 → 2025

Which districts saw the biggest new property price surge over the last nine years?

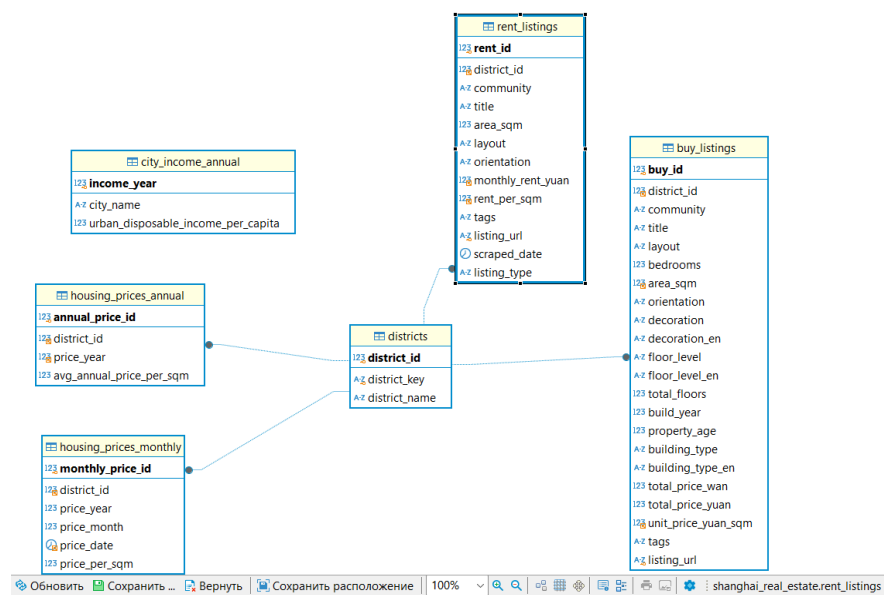


Price-to-Income Ratio Trend

How many years of income (take-home pay) are needed to purchase 1 sqm of housing in Shanghai throughout the years (2016–2025)?



C. System Implementation



```

(venv) PS C:\Users\korepanova\Desktop\shhomework3> python .\02_import_csv.py
Buy listings loaded
Rent listings loaded
Monthly prices loaded
Annual prices loaded
Income data loaded

Rows in database:
districts: 16
buy_listings: 2084
rent_listings: 1852
housing_prices_monthly: 2020
housing_prices_annual: 175
city_income_annual: 11
  
```



```

Командная строка  X Windows PowerShell  X + -
1 - Show all districts
2 - Find sale listings
3 - Find rental listings
4 - Find rental listings by tag
5 - Search by community name
6 - Show annual prices
7 - Show monthly prices
8 - Show city income
9 - Find a listing by URL
0 - Exit
Choose an option: 2
District name: pudong
Bedrooms: 2
Minimum area: 50
Maximum total price: 5000000
Listings found: 44
(809, '金地未来', '精装修保养好, 高层楼层采光好, 看房方便', 2, Decimal('65.00'), Decimal('3980000.00'), Decimal('61231.00'), 'https://sh.lianjia.com/ershoufang/107115149158.html')
(810, '壹唐八村', '店长推荐高得房率, CBD位置, 全明户型', 2, Decimal('61.92'), Decimal('1660000.00'), Decimal('26809.00'), 'https://sh.lianjia.com/ershoufang/107115441206.html')
(812, '高丰北苑', '18年次新房 高区景观楼层 三开间朝南 诚意出售 有钥匙', 2, Decimal('85.61'), Decimal('3350000.00'), Decimal('39131.00'), 'https://sh.lianjia.com/ershoufang/107115545122.html')
(815, '青山新村西北街坊', '近地铁, 双南两房, 房东急售急售急售', 2, Decimal('60.45'), Decimal('2888000.00'), Decimal('47789.00'), 'https://sh.lianjia.com/ershoufang/107115627148.html')
(817, '三林世博家园(东书房路390弄)', '近地铁6, 11, 13号线, 小区位置好楼层好, 看房方便', 2, Decimal('73.16'), Decimal('3250000.00'), Decimal('44424.00'), 'https://sh.lianjia.com/ershoufang/107115486636.html')
(822, '瀚盛家园', '张江东+唐镇国际社区+2号线+刚需盘+精装修+好户型', 2, Decimal('87.15'), Decimal('2880000.00'), Decimal('33047.00'), 'https://sh.lianjia.com/ershoufang/107115294673.html')
(823, '金丰城市广场', '次新房 低总价 刚需好上车 南北通 采光视野好', 2, Decimal('79.54'), Decimal('2290000.00'), Decimal('28791.00'), 'https://sh.lianjia.com/ershoufang/107115334623.html')
(827, '恒大新城新苑(西区)', '近轨通, 南北两房一厅6-8-11三轨相交, 采光好 楼层好', 2, Decimal('68.90'), Decimal('3450000.00'), Decimal('50073.00'), 'https://sh.lianjia.com/ershoufang/107115180609.html')
(830, '妙境一村', '近地铁, 出行便利, 户型方正, 南北格局通透大两房', 2, Decimal('80.54'), Decimal('2180000.00'), Decimal('27068.00'), 'https://sh.lianjia.com/ershoufang/107115263558.html')
(837, '瑞泰华庭', '山姆步行可达, 地铁700米, 南北格局采光通风佳', 2, Decimal('89.40'), Decimal('3750000.00'), Decimal('41947.00'), 'https://sh.lianjia.com/ershoufang/107114733027.html')
(838, '万泰佳园', '此房满五 南北通 全明户型 采光通风好 中间楼层 位置', 2, Decimal('77.00'), Decimal('2780000.00'), Decimal('36104.00'), 'https://sh.lianjia.com/ershoufang/107115629097.html')
(840, '瑞源四村', '近前CBD, 开发商样板房 少有弧形阳台采光充足, 直接入住', 2, Decimal('77.71'), Decimal('1890000.00'), Decimal('24322.00'), 'https://sh.lianjia.com/ershoufang/107115341220.html')

```

```

1 - Show all districts
2 - Find sale listings
3 - Find rental listings
4 - Find rental listings near metro
5 - Search by community name
6 - Show annual prices
7 - Show monthly prices
8 - Show city income
9 - Find a listing by URL
0 - Exit
Choose an option: 7
District name: pudong
Year: 2025
Rows found: 12
(2025, 1, datetime.date(2025, 1, 1), Decimal('69128.00'))
(2025, 2, datetime.date(2025, 2, 1), Decimal('68141.00'))
(2025, 3, datetime.date(2025, 3, 1), Decimal('67300.00'))
(2025, 4, datetime.date(2025, 4, 1), Decimal('69200.00'))
(2025, 5, datetime.date(2025, 5, 1), Decimal('65000.00'))
(2025, 6, datetime.date(2025, 6, 1), Decimal('64900.00'))
(2025, 7, datetime.date(2025, 7, 1), Decimal('63800.00'))
(2025, 8, datetime.date(2025, 8, 1), Decimal('64000.00'))
(2025, 9, datetime.date(2025, 9, 1), Decimal('66400.00'))
(2025, 10, datetime.date(2025, 10, 1), Decimal('66000.00'))
(2025, 11, datetime.date(2025, 11, 1), Decimal('66000.00'))
(2025, 12, datetime.date(2025, 12, 1), Decimal('66400.00'))

```